
Development and Implementation of a Loyalty System for a Confectionery Based on Smart Contracts

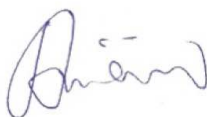
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JUNE 2026
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Abstract

This diploma project presents the design and MVP-oriented implementation plan of a smart-contract-based loyalty system for a confectionery ecosystem. The problem addressed is the fragmentation of conventional loyalty programs, where reward points are stored in isolated databases, have limited cross-partner utility, and provide insufficient transparency for issuance, redemption, and settlement.

The objective is to develop an architecture for a tokenized loyalty platform combining blockchain-based auditability with low-friction user interaction. The proposed solution uses TON blockchain for settlement-critical events, Telegram Mini App as the interaction layer, Node.js and Express.js for backend services, and PostgreSQL for off-chain operational data. The work follows Design Science Research: literature review and system comparison are transformed into requirements, architecture models, UML/BPMN diagrams, interface-flow specifications, pseudocode-level logic, and architectural decision records.

The main result is a reproducible MVP blueprint for a hybrid on-chain/off-chain loyalty platform supporting token issuance, redemption, configurable reward rules, POS/e-commerce integration, blockchain transaction recording, analytics, and partner participation. Evaluation covers security, transparency, availability, scalability, performance, controlled pre/post demonstration indicators, and business KPI estimates. Results show substantial improvement: transparency satisfaction increased from 1.79 to 4.38 (out of 5), lost-points concern decreased from 68.1% to 10.6%, and the share of businesses expecting excessive manual reconciliation workload decreased from 65.2% to 13.0%. The project concludes that the TON + Telegram Mini App approach is feasible at the architecture and controlled-demonstration level, while full operational validation requires pilot deployment, performance testing, smart-contract security audit, and longitudinal user-engagement analysis.

Keywords: blockchain loyalty system, smart contracts, TON, Telegram Mini App, confectionery, tokenized rewards, Design Science Research, hybrid architecture.

Dedication and acknowledgements

This diploma project represents the culmination of our academic journey at Astana IT University, and it would not have been possible without the support and guidance of many individuals.

First and foremost, we would like to express our sincere gratitude to our academic supervisor for their invaluable guidance, continuous encouragement, and insightful feedback throughout the entire research and development process. Their expertise in distributed systems and rigorous approach to academic research profoundly shaped the quality of this work.

We extend our deep appreciation to the faculty members of Astana IT University for providing a strong educational foundation and fostering an environment of intellectual curiosity and technological innovation.

We also wish to thank the participating confectionery businesses and customers who generously provided their time and feedback during our preliminary surveys and MVP demonstrations. Your real-world insights were instrumental in validating our architectural decisions.

Finally, we dedicate this work to our parents, families, and friends. Thank you for your unwavering patience, emotional support, and belief in our abilities throughout our university studies. Your constant encouragement has been our greatest motivation.

Definitions

Following terms are used in this work:

Blockchain loyalty system	A loyalty platform in which critical reward operations (issuance, transfer, redemption) are recorded and verified through blockchain infrastructure.
Coalition loyalty model	A multi-partner loyalty structure where several independent businesses issue and accept rewards under shared rules.
Smart contract	A programmable on-chain component that executes pre-defined business logic automatically when specified conditions are met.
Tokenized reward unit	A digital loyalty value represented as a blockchain token and used for accumulation, transfer, or redemption.
Soulbound token (SBT)	A non-transferable token used as a persistent credential for access control and partner/business identity confirmation.
On-chain data	Transaction-critical information persisted in blockchain state, providing immutability, traceability, and non-repudiation.
Off-chain data	Operational or personal data stored outside the blockchain (e.g., in relational databases) to improve privacy and performance.
Hybrid data architecture	A storage and processing approach combining on-chain trust anchors with off-chain operational data management.

Auditability	The ability to reconstruct and verify the full lifecycle of loyalty events from user action to final recorded state.
Interoperability	The ability of different partner systems and services to exchange, interpret, and process loyalty-related data consistently.
MVP (Minimum Viable Product)	The minimal implementation scope required to validate architecture feasibility and core business assumptions.
CAP theorem	A fundamental result in distributed systems stating that a networked shared-data system can guarantee at most two of the following three properties simultaneously: Consistency, Availability, and Partition tolerance.
Eventual consistency	A consistency model in which, given no new updates, all replicas of a data item will eventually converge to the same value; used in systems that prioritize availability over immediate consistency.
Actor model	A concurrency paradigm in which independent computational entities (actors) communicate exclusively via asynchronous message passing, avoiding shared mutable state.

Designations and Abbreviations

Following designations and abbreviations are used in this work:

API	Application Programming Interface.
B2B	Business-to-Business interaction model.
BPMN	Business Process Model and Notation.
C4	Context, Containers, Components, and Code architecture model.
DSR	Design Science Research methodology.
EDA	Event-Driven Architecture.
FR	Functional Requirement.
JWT	JSON Web Token used for stateless authentication.
MFA	Multi-Factor Authentication.
MVP	Minimum Viable Product.
NFR	Non-Functional Requirement.
PII	Personally Identifiable Information.
POS	Point of Sale system.
RBAC	Role-Based Access Control.
SBT	Soulbound Token (non-transferable token credential).
SDK	Software Development Kit.
SQL	Structured Query Language.
TMA	Telegram Mini App.
TON	The Open Network blockchain ecosystem.
UML	Unified Modeling Language.

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Introduction

Work relevance. Loyalty management is undergoing structural transformation under the pressure of digital competition, omnichannel customer behavior, and increasing expectations for transparent value exchange. Early work on privacy-preserving electronic loyalty systems already identified the tension between personalization and data protection in digital marketplaces [5]. In B2B-oriented confectionery distribution, conventional loyalty systems remain fragmented across isolated applications and closed point balances, which weakens user motivation and reduces actual redemption intensity [6], [1]. The confectionery value chain requires interoperable incentive logic, reliable partner accounting, and predictable reconciliation rules.

Problem statement. The central practical problem addressed in this work is the contradiction between the economic promise of multi-brand loyalty ecosystems and the operational inefficiency of conventional implementations. Traditional points infrastructures are characterized by delayed transferability, limited cross-brand utility, and high administrative overhead caused by manual fraud-control procedures. Preliminary stakeholder input also indicates that fragmented program rules and perceived loss of points hinder loyalty participation.

Blockchain technology offers a potential resolution by enabling immutable transaction histories and programmable reward logic through smart contracts [3]. Distributed ledger technology (DLT) has already demonstrated commercial viability across supply-chain, financial, and identity-management domains [7], while systematic reviews confirm growing research interest in blockchain-based business applications beyond cryptocurrency [8]. However, implementation complexity can shift friction to end-user onboarding if participants must manage wallets and gas fees. Thus, the challenge is to integrate blockchain while preserving accessibility for non-crypto-native stakeholders.

Goal of the work. The primary objective of this research is to design and conceptually validate a high-performance B2B confectionery loyalty ecosystem based on TON blockchain infrastructure and Telegram Mini App interaction flows. The intended system

transforms loyalty points from isolated database counters into tokenized digital assets governed by programmable exchange and redemption logic.

Research object and subject. The research object is a blockchain-enabled loyalty artifact for B2B confectionery ecosystems. The research subject is the set of architectural, methodological, and governance mechanisms that determine whether such an artifact can provide practical operational value, specifically interaction between smart-contract reward logic, mini-app interfaces, and partner-level settlement processes.

Scientific novelty. The scientific novelty of the work lies in integrating a high-throughput blockchain architecture with social-native interaction infrastructure to model a practically deployable B2B “gas-light” loyalty flow where end-user friction is minimized through relayer-based interaction design [1]. This research proposes a modular multi-brand framework adapted to coalition conditions and aligned with measurable design criteria.

Practical relevance of the work. The expected utility of the proposed approach is three-fold. First, smart-contract automation can reduce manual reconciliation effort and improve traceability of reward issuance/redemption events. Second, coalition-compatible token logic supports cross-brand incentive circulation. Third, interface delivery through Telegram Mini Apps decreases onboarding barriers by avoiding context switches to unfamiliar standalone tools. Together, these effects improve adoption feasibility for medium-sized business ecosystems where implementation budget and operational continuity are critical [9].

Objectives:

1. **RQ1:** Analyze the economic, behavioral, and technical barriers to B2B loyalty participation in fragmented confectionery ecosystems [6].
2. Synthesize and compare existing blockchain-based loyalty approaches, identifying transferable and non-transferable design patterns [10], [2].
3. **RQ2:** Design a decentralized architecture on TON that enables cross-brand token exchange and auditable settlement logic [3], [1].
4. Define a low-friction interaction model using Telegram Mini Apps and relayer-assisted transaction execution [1].
5. **RQ3:** Prepare an MVP-level system specification (requirements, architecture, UML/BPMN models, and interface flows) and define evaluation metrics to validate whether baseline feasibility requirements are satisfied.

Structure of the work. Chapter 1 presents the literature review, analysis of existing systems, technology comparison, and formalized research gap. Chapter 2 describes the methodology, data collection logic, and evaluation criteria. Chapter 3 provides MVP-oriented system design artifacts: requirements, architecture, UML/BPMN diagrams, interface flows, and economic-efficiency assumptions. Chapter 4 presents the results and discussion, including pre/post demonstration indicators and business KPI evaluation. The conclusion summarizes key findings and next-stage research priorities.

Chapter 1

Literature Review

1.1 Blockchain-based loyalty systems: concepts and models

At their core, blockchain loyalty systems force a shift in how trust is managed: rather than relying on a single institution's database, rewards become programmable assets verified at the protocol level [6], [1], [2], [3]. Interestingly, the drive toward privacy-preserving rewards actually predates the current blockchain hype. For example, Enzmann and Schneider [5] outlined an early architecture for electronic marketplaces that strictly prioritized user-data minimization long before distributed ledgers became mainstream. Despite the evolving technology stack, the underlying mechanics of these programs consistently follow a standard six-stage lifecycle across the literature, which provides a useful baseline for modeling system interactions [2], [9], [10].

Most implementations can be decomposed into four interacting layers: interaction, application/service, smart-contract, and infrastructure [9], [2], [3]. Figure 1.2 summarizes this stable layer composition.

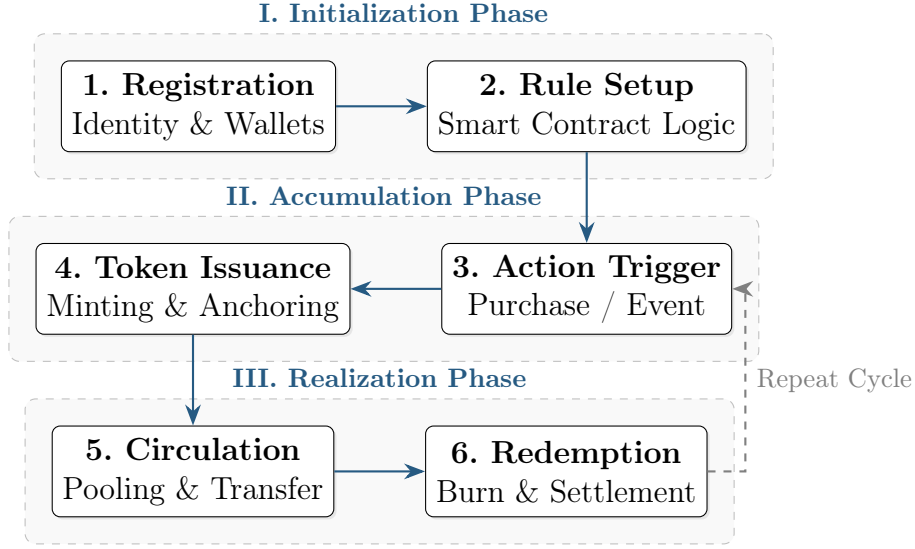


Figure 1.1: Blockchain-based loyalty lifecycle model synthesized from prior implementations and conceptual studies [1], [2], [3].

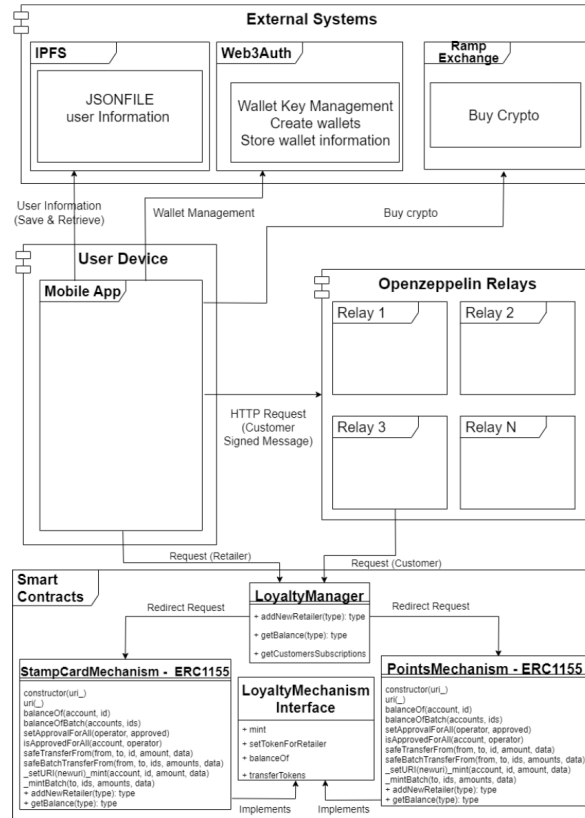


Figure 1.2: Reference architecture pattern for blockchain loyalty platforms.

1.2 Analysis of existing systems

Blockchain-native proposals typically try to bypass the limitations of centralized databases by simply moving the critical settlement logic directly into smart contracts. In theory, this makes issuance, transfer, and redemption completely transparent [1], [3]. However, as Konstantinidis et al. [8] point out in their systematic review, while everyone is talking about using blockchain for business applications, actual real-world deployments in the loyalty sector are still surprisingly rare. This gap strongly reinforces the need for research that focuses on actual implementation hurdles rather than just high-level concepts.

The prototype discussed by Sönmeztürk et al. demonstrates the token flow logic between manufacturers, markets, and customers (see Figure 1.3 and Figure 1.4) [1].

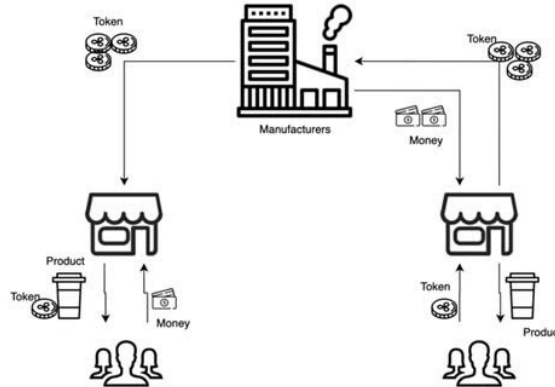


Figure 1.3: TECH token flow logic in the system (adapted from Sönmeztürk et al., Figure 4 [1]).

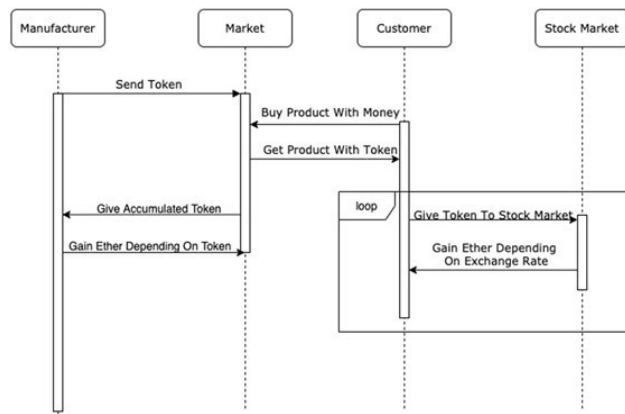


Figure 1.4: Time diagram of token flow (adapted from Sönmeztürk et al., Figure 5 [1]).

Furthermore, Heidari (2025) formalizes a tokenized incentive mechanism through an S–O–R (Stimulus–Organism–Response) simulation model, bridging token economics with measurable engagement behavior (Figure 1.5) [4].

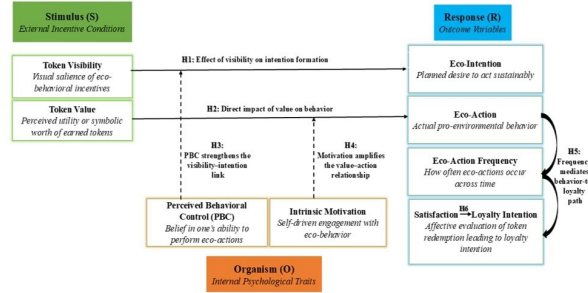


Figure 1.5: S–O–R simulation model of tokenized smart-contract incentives (adapted from Heidari, 2025, Figure 1 [4]).

1.2.1 Related Work Comparison

To highlight the precise technical and methodological gap this project fills, Table 1.1 explicitly compares the proposed Sweet Loyalty architecture against prominent frameworks from the recent literature.

The comparison focuses on implementation-relevant dimensions: blockchain access model, execution paradigm, user interaction, gas-fee handling, privacy boundary, and token lifecycle. These criteria are selected because they determine whether a loyalty platform can move from a conceptual blockchain proposal to a deployable retail workflow.

Table 1.1: Comparison of the proposed architecture against existing blockchain loyalty frameworks.

Feature	Sönmeztürk et al. [1]	Nguyen et al. [3]	Tu et al. [11]	Proposed MVP (Sweet Loyalty)
Blockchain	Ethereum (Public)	Hyperledger (Consortium)	Hyperledger Fabric	TON (Public)
Execution Model	Synchronous (EVM)	Synchronous	Synchronous	Asynchronous (Actor Model)
User Interaction	Web3 Wallet (High friction)	Enterprise portal	App/Web portal	Telegram Mini App (Low friction)
Gas Fees	Paid by user	Abstracted (Consortium)	Abstracted (Consortium)	Paid by Relayer (Gasless for user)
Data Privacy	On-chain balances	Private Channels	Private Channels	Hybrid (Off-chain PII, On-chain settlement)
Token Lifecycle	Standard ERC-20	Cross-brand exchange	Call Auction market	TEP-74 with verifiable burn

1.2.2 Taxonomy of Blockchain Loyalty Architectures

To systematically map the research landscape and identify the academic gap, existing proposals were categorized along two critical dimensions: the **ledger access model** (Public vs. Consortium vs. Private) and the **smart-contract execution paradigm** (EVM-synchronous vs. UTXO vs. Actor-model asynchronous).

Table 1.2 presents this taxonomy. It reveals that while substantial research exists on EVM-based consortium models [10], [3], and Hyperledger Fabric-based coalition systems with peer-to-peer point exchange have been explored [11], there is a distinct gap in utilizing asynchronous actor-model public blockchains (like TON) combined with off-chain aggregation for high-frequency retail loyalty.

Table 1.2: Taxonomy of blockchain loyalty architectures in recent literature.

Execution Paradigm	Public Permissionless	Consortium / Federated	Private Permissioned
EVM (Synchronous)	High cost, low throughput. Focus of early conceptual works (e.g., ERC-20 loyalty tokens).	Dominant paradigm. Hyperledger Besu/Quorum setups. High governance overhead [3].	Enterprise Ethereum. High privacy, but lacks open composability.
UTXO	Rare for complex logic, mostly simple asset issuance (e.g., Colored Coins).	Sometimes used in Corda-based financial settlements, rarely for retail loyalty.	High privacy, complex state management.
Actor-Model (Asynchronous)	Proposed MVP (TON). Highly scalable, micro-transaction viable, relies on eventual consistency.	Unexplored in loyalty literature.	Unexplored in loyalty literature.

1.2.3 Production blockchain-loyalty cases and coalition network effects

Production-oriented blockchain loyalty initiatives show that technical novelty alone is not sufficient for sustainable adoption. Singapore Airlines announced KrisFlyer as a blockchain-based airline loyalty wallet that lets members convert miles into spendable value with merchant partners, demonstrating that blockchain can support closed-loop reward settlement when it is embedded into an existing high-trust brand and partner network [12]. Starbucks Odyssey followed a different Web3 strategy: it extended Starbucks Rewards with digital collectible assets and experiential benefits [13]. However, the programme was later discontinued in 2024, illustrating that branded NFT mechanics can fail to create durable utility if the reward layer is perceived as separate from everyday purchase behavior [14]. These cases reinforce the design requirement that the blockchain layer must remain operationally useful and mostly invisible to non-technical users.

Coalition loyalty research further suggests that the main value of multi-brand rewards comes from network effects rather than from point issuance alone. Coalition membership can create indirect effects when a new merchant increases the utility of the whole reward currency, but it can also generate free-riding, liability-allocation, and cross-merchant spillover risks [15], [16], [17]. For the confectionery context, this means that

smart contracts must not only tokenize rewards; they must also make partner eligibility, settlement evidence, and redemption rules auditable. The proposed TON + Telegram Mini App architecture therefore prioritizes low-friction access, partner governance, and off-chain privacy over speculative secondary trading.

1.3 Technology comparison and smart contract paradigms

To select an appropriate technology stack for the MVP, candidate platforms were compared using criteria relevant for B2B coalition networks: operational economics, UX complexity, and scalability [7], [8]. A critical dimension of this comparison is the underlying smart-contract execution paradigm. Traditional EVM-based systems operate synchronously: all contract calls within a transaction execute atomically, halting the global state until completion. While this simplifies development, it severely limits throughput and creates gas-fee volatility during network congestion, rendering EVM unviable for micro-transaction retail environments [10], [3]. Tu et al. [11] demonstrate an alternative using Hyperledger Fabric with a call-auction exchange mechanism, achieving cross-organizational point circulation; however, their consortium-permissioned approach introduces governance overhead and lacks the open composability offered by public chains.

In contrast, the TON (The Open Network) blockchain employs an asynchronous actor model [18]. Smart contracts on TON communicate exclusively via asynchronous message passing. This paradigm prevents global state bottlenecks, enabling near-infinite sharding and massive parallel execution. Consequently, loyalty tokens implemented on TON (using the TEP-74 standard) require a different architectural mindset: operations such as token transfers or balances checks do not return immediate synchronous responses. The architecture must handle eventual consistency, race conditions, and message-bounce handling (action phases).

Table 1.3 summarizes the broader ecosystem evaluation.

Based on this comparison, the thesis proceeds with TON and Telegram Mini App integration as the foundation, prioritizing user accessibility and high throughput required for retail transaction volumes.

Table 1.3: Technology comparison matrix for coalition loyalty MVP architecture.

Technology family	Main advantages	Main constraints	Fit for thesis context
Ethereum / EVM public networks	Strong tooling maturity, reusable token standards, broad ecosystem support	Potential fee volatility and higher UX friction for non-crypto-native users	Medium-to-high fit for technical rigor; medium fit for low-friction onboarding
EVM Layer-2 / sidechain stack	Lower transaction costs, better throughput profile, contract portability	Additional architectural complexity around bridging and finality assumptions	High fit for cost-sensitive MVPs if operations are abstracted well
TON + Telegram-oriented integration	Channel alignment with Mini App strategy and rapid interaction flows	Ecosystem-specific decisions require careful modularization	High fit for interaction-driven B2B adoption in this thesis
Permissioned consortium ledger	Strong role control, predictable execution, easier policy containment	Reduced openness/composability and governance overhead	Medium-to-high fit where the partner set is closed and centrally coordinated

1.4 Research gap and problem statement

Most blockchain-loyalty studies emphasize protocol-level feasibility but rarely provide an implementation-ready architecture addressing B2B coalition governance and onboarding friction [10], [19], [2]. This study addresses this gap by proposing a “gas-light” relay-assisted loyalty architecture that balances technical transparency with operational simplicity, directly motivating the system design in Chapter 3.

Chapter 2

Methodology of the Work

The study’s methodology is based on Design Science Research (DSR), focusing on the practical design and evaluation of an IT artifact for confectionery ecosystems.

2.1 Data collection and survey instrument design

Data collection encompasses handling system data and empirical input gathered via structured surveys (Appendix A). To ensure construct validity and statistical reliability, the survey instrument was deployed across two distinct cohorts: a B2C customer group ($N = 47$) and a B2B partner group ($N = 23$). Respondents were selected using purposive sampling within the urban centers of Astana and Almaty to capture relevant market behavior.

The questionnaire utilized a 5-point Likert scale to measure constructs such as transparency satisfaction, perceived usefulness, and trust. To evaluate the exact impact of the artifact, a pre-test/post-test experimental design was applied. Participants evaluated traditional systems (pre-demonstration phase), interacted with the proposed MVP, and then re-evaluated the concepts (post-demonstration phase).

Core system data includes partner profiles, token balances, SBT records, cashback transactions, and audit logs. The system uses a hybrid architecture: operational data (PostgreSQL) is stored off-chain, while loyalty-value state transitions are anchored on-chain [20], [21].

Data enters via Telegram Mini App, POS actions, admin panel, and TON events.

Incoming payloads undergo schema-level validation before persistence, while blockchain-bound operations are executed via adapters and reconciled off-chain.

Table 2.1: System data inventory, source, format, storage, and architecture linkage in the Sweet Loyalty MVP.

Data type	Source	Format / storage	Processing / functionality
Partner profile	Telegram Mini App form; admin input	JSON/API payload and SQL row; PostgreSQL off-chain storage	Auth and partner-management services; registration, RBAC, and partner status control
SWEET token balance and transfer state	TON blockchain events; TON API reads	TON transaction/event structures and API JSON; source of truth on TON with PostgreSQL cache	Blockchain adapter and loyalty service; issuance accounting, transfers, and redemption checks
SBT certificate ownership	Admin panel minting; TON contract state	TON NFT/SBT metadata and ownership records; stored on TON blockchain	Blockchain adapter and access-control logic; B2B access gating and business identity proof
Cashback transaction record	POS-triggered purchase event; API request	JSON request with transaction reference; PostgreSQL record with TON reference hash	Ecosystem/loyalty service and API layer; FR1/FR2 purchase-to-reward lifecycle
Reward claim and catalog usage	User redemption action; admin catalog update	JSON/API payload and SQL row; PostgreSQL with on-chain settlement reference	Loyalty and partner services; reward redemption and partner-side reconciliation
Audit and admin action log	Admin panel actions; backend system events	Structured log/event records; PostgreSQL/log storage off-chain	Admin/audit service; traceability, incident review, and governance accountability

Table 2.2: MVP implementation status of key project data categories.

Data category	Primary source	Storage pattern	MVP status
Partner data	Telegram Mini App forms	PostgreSQL off-chain	Implemented as real backend data flow
Jetton balance/state	TON API and TON events	On-chain with off-chain cache	Implemented through real on-chain reads
SBT access records	Admin panel mint action	On-chain ownership state	Implemented as real on-chain state
Cashback transaction trail	POS/business transaction trigger	Hybrid: blockchain reference and database record	Implemented as hybrid flow
Analytics and achievement data	Derived from system events	Planned aggregation layer	Planned/mock in current MVP stage

This collection design balances privacy, performance, and transparency. Off-chain storage preserves privacy, while on-chain anchoring ensures non-repudiation of loyalty

transitions.

2.2 Research design and evaluation metrics

The DSR workflow entails problem diagnosis, requirement definition, architecture design, and evaluation [22]:

1. **Problem identification:** Conventional loyalty systems lack interoperability and auditability [1].
2. **Requirement elicitation:** Derived from literature, interviews, and market comparisons.
3. **Artifact design:** A layered hybrid on-chain/off-chain architecture [20].
4. **System modeling:** UML, BPMN, and C4 models describe structure and behavior.
5. **MVP specification:** Focuses on validating transparent reward lifecycles and partner interoperability.
6. **Evaluation:** Based on ATAM and quality-attribute scenarios (security, transparency, availability, scalability, performance) [23].

Table 2.3: System evaluation metrics and conceptual measurement criteria used in architecture assessment.

Metric	Evaluation indicator	Measurement criterion
Security	Policy enforcement and transaction integrity	Restricted actions execute only under authorized roles; invalid operations do not alter ledger state
Transparency	Auditability of reward lifecycle events	Business-critical actions are traceable from user action to immutable transaction evidence
Availability	Service continuity during failure and load events	Core user operations remain available through failover and retry strategies
Scalability	Ability to support growth in users and partners	Application services can scale horizontally without redesign of the smart-contract core
Performance	Operational latency in customer-facing workflows	Issuance and redemption flows remain responsive through asynchronous handling of confirmation delays

Representative stimulus–response scenarios include: unauthorized token issuance attempts (security scenario), customer requests for verifiable redemption history (transparency scenario), temporary service-node failure at campaign peak (availability scenario), onboarding of new partner adapters without core contract redesign (scalability scenario), and temporary blockchain confirmation delay with asynchronous status update (performance scenario).

Trade-offs (e.g., transparency vs. performance) are mitigated through selective on-chain recording and asynchronous workflows.

2.2.1 Research propositions and evaluation criteria

Based on the literature review and the identified research gap, the following directional propositions are used to structure the pre/post demonstration evaluation:

- P1** Exposure to the MVP demonstration increases customer-perceived transparency satisfaction (measured as mean Likert score on construct C4).
- P2** Exposure to the MVP demonstration reduces customer concern about lost or unaccounted reward points (measured as proportion reporting “Yes” to point-loss worry).
- P3** Post-demonstration, business respondents report lower expected manual reconciliation workload (measured as proportion reporting >3 hours/week).
- P4** Post-demonstration, partner interest in a shared blockchain-enabled loyalty platform increases (measured as proportion scoring 4–5 on the adoption-intention item).

P1 and P2 address customer-side value perception and risk reduction, while P3 and P4 address partner-side operational efficiency and adoption intent. These propositions are evaluated using paired-samples tests (P1–P2, where $N_{B2C} = 47$) and descriptive effect-size analysis (P3–P4, where $N_{B2B} = 23$). Effect sizes are reported as Cohen’s d for continuous means and Cohen’s h for proportions, following standard practice for pre/post designs with moderate sample sizes.

2.3 Evaluation data processing and interpretation

The MVP is evaluated using pre- and post-demonstration surveys (47 B2C and 23 B2B respondents) and a business KPI model. Data processing involves converting raw Likert responses into mean scores and positive-response rates, and translating business categories into KPI estimates (transaction volume, administration cost, platform fee, payback).

Table 2.4: Evaluation dataset, processing logic, and interpretation method.

Data block	Selection logic	Processing method	Interpretation
B2C pre/post survey	Customer perception of loyalty transparency, Telegram access, trust, and cross-partner utility	Mean Likert scores; positive-response rate for scores 4–5; before/after comparison	Indicates whether the MVP improves perceived customer value and reduces point-loss concern
B2B pre/post survey	Partner interest, campaign control, dispute reduction, and reconciliation workload	Percentage distributions; percentage-point change; selected Likert positive-response rates	Indicates whether the MVP reduces partner-side operational friction and improves adoption intention
Business KPI model	Transaction volume, basket size, administration cost, platform fee, and payback assumptions	Midpoint conversion of survey categories; net monthly benefit and payback formulas	Estimates design-stage business feasibility and identifies metrics for pilot validation
Architecture acceptance criteria	Security, transparency, availability, scalability, and performance requirements	Scenario-based assessment and requirement-to-component traceability	Evaluates whether the artifact is coherent enough to proceed to pilot deployment

Reliability is supported by the fact that all transformations are explicit and repeatable: the same raw responses produce the same descriptive indicators and KPI estimates. Validity is supported by traceability between measured indicators and system requirements. For example, the transparency score evaluates whether NFR7 is meaningful to users; reconciliation workload evaluates whether partner settlement automation has operational value; and net monthly benefit evaluates whether the application can create a business case beyond technical novelty.

Table 2.5: Methodology validity map linking research questions, evidence sources, processing logic, and validity controls.

RQ	Data source	Metric / evidence	Processing	Threat to validity	Mitigation
RQ1	Literature review; B2C/B2B baseline survey	Fragmentation barriers, point-loss concern, manual reconciliation workload	Thematic synthesis and descriptive baseline rates	Sample and novelty bias	Triangulation with prior studies and explicit limitation of generalizability
RQ2	Architecture models, requirements, ADRs, technology comparison	Security, transparency, interoperability, scalability, performance criteria	Requirement-to-component mapping and scenario-based assessment	Design bias toward selected stack	Comparison with centralized, EVM, TON, and permissioned alternatives
RQ3	Controlled pre/post demonstration; TON sandbox tests; KPI model	Pre/post change, effect size, gas fee, execution time, net benefit, payback	Mean/proportion comparison, Cohen’s d/h , deterministic benchmark, formula-based KPI estimation	Controlled setting cannot prove production adoption	Results framed as architecture and demonstration-level feasibility; pilot validation defined as future work

2.4 Validity, limitations and ethical considerations

Research rigor is strengthened through methodological triangulation, established architecture evaluation (ATAM), standardized modeling (UML, BPMN), and explicit traceability. Limitations include the conceptual MVP scope, limited sample sizes, and lack of long-term retention data. Ethical considerations focus on privacy (PII kept off-chain) and fairness in automated loyalty rules.

Internal validity is addressed by keeping the demonstration sequence consistent for both respondent groups: participants first evaluated the conventional loyalty context, then reviewed the same MVP concept and evidence flow, and finally completed the post-demonstration items. This does not eliminate novelty bias, but it makes the before/after comparison interpretable because the intervention is clearly defined. Construct validity is supported by linking each measured indicator to a concrete system requirement: transparency satisfaction corresponds to auditable reward history, lost-points concern corresponds to immutable balance evidence, and reconciliation workload corresponds to partner-side settlement automation.

Reliability is supported through repeatable processing rules. Likert-scale responses are transformed into mean scores and positive-response rates using explicit thresholds, while business KPI estimates are calculated from documented assumptions rather than subjective judgement. The same source files and formulas can therefore reproduce the reported tables and figures. External validity remains limited because the evaluation reflects a controlled MVP demonstration rather than a longitudinal market deployment; for this reason, the results are interpreted as feasibility and adoption-readiness evidence, not as proof of production-scale market performance.

The ethical boundary of the design is especially important because loyalty systems can influence purchasing behavior and process sensitive customer information. The proposed architecture avoids storing personal identifiers or purchase-level details directly on-chain, where removal or correction would be impractical. Automated loyalty rules are also treated as configurable business policies rather than hidden scoring mechanisms: partner eligibility, campaign parameters, and redemption constraints must remain explainable to both customers and business participants. These safeguards reduce the risk that blockchain transparency is achieved at the expense of customer privacy or unfair partner treatment.

Table 2.6: Limitations of the study, expected impact, and mitigation approach at the current research stage.

Limitation	Impact on findings	Mitigation approach
MVP-level scope without production rollout	Limits claims on real-world scale and long-term operation	Focus on architecture feasibility, explicit assumptions, and scenario-based validation
Limited interview sample	Constrains generalization across broader business populations	Triangulate stakeholder insights with literature and market benchmarking
No long-term behavioral observations	Prevents evaluation of sustained loyalty and retention effects	Position results as design-stage evidence and define longitudinal evaluation as future work
Blockchain and integration technical uncertainty	May affect runtime reliability and implementation outcomes	Include risk-aware architecture choices and document unresolved risks transparently
Limited regional causal data on blockchain loyalty	Repeat-purchase uplift cannot be claimed as an observed Central Asian effect	Treat uplift as a conservative scenario variable and validate it in a live pilot

Chapter 3

MVP and System Design

3.1 Requirements (functional/non-functional)

This section formalizes the functional requirements (FR) and non-functional requirements (NFR) for the proposed MVP, based on the identified problem statement and coalition constraints.

Functional Requirements (FR)

Table 3.1: Functional requirements for the proposed loyalty platform MVP.

ID	Requirement
FR1	The system shall issue loyalty tokens automatically after successful purchase confirmation.
FR2	The system shall allow customers to redeem loyalty tokens during checkout.
FR3	The system shall allow administrators to configure reward rules without smart-contract redeployment.
FR4	The system shall integrate with POS and e-commerce systems through API interfaces.
FR5	The system shall record all token-related transactions on the blockchain ledger.
FR6	The system shall provide dashboards for analytics and campaign-performance monitoring.
FR7	The system shall support partner merchants in accepting tokens for redemption.

Functional requirements explanation. FR1-FR2 represent the core value cycle (issuance and redemption). FR3-FR7 define critical governance, interoperability, reporting, and operational components necessary for a viable ecosystem.

Non-Functional Requirements (NFR)

Table 3.2: Non-functional requirements and target operational measures.

ID	Requirement	Target measure
NFR1	Availability	At least 99.5% uptime during business hours
NFR2	Performance	POS transaction latency below 2 seconds
NFR3	Security	All administrative operations require RBAC and MFA
NFR4	Privacy	Personally identifiable information (PII) shall never be stored on-chain
NFR5	Scalability	The system shall handle 10× peak seasonal load
NFR6	Reliability	No loss of issued tokens shall occur under system failure
NFR7	Transparency	Customers shall be able to audit their transaction history

Non-functional requirements explanation. NFR1-NFR2 ensure checkout business continuity. NFR3-NFR4 handle compliance and data privacy. NFR5 manages load spikes, while NFR6-NFR7 safeguard asset integrity and foster user trust through transparency.

3.2 Architecture of the project

Figure 3.1 presents the system-context view, defining external actors, interaction channels, and the external blockchain integrity layer. This C4-model context diagram establishes the boundary between the internal platform components and the external environment. The primary actors include retail customers (who interact via the Telegram Mini App), partner merchants (who interact via the web-based admin panel), and external POS systems (which trigger cashback issuance via secure REST APIs). The TON blockchain acts as a decentralized trust anchor, operating outside the central administrative domain, ensuring that reward issuance and redemption events are cryptographically verified and globally auditable by all coalition participants.

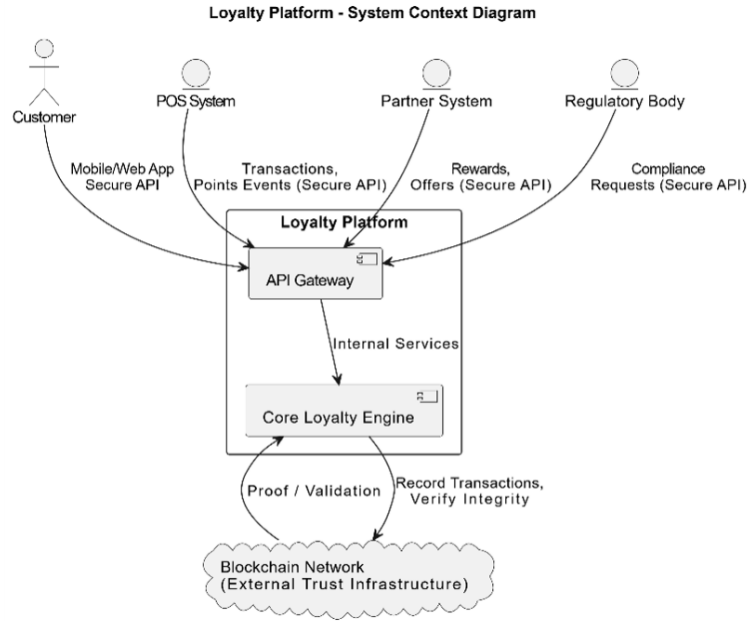


Figure 3.1: Confectionery system context diagram for the proposed loyalty platform.

Figure 3.2 illustrates the high-level architecture across presentation, application, blockchain, and data layers. The presentation layer abstracts blockchain complexity through Telegram’s native UI, while the application layer orchestrates business logic via Node.js microservices. The blockchain layer consists of FunC smart contracts responsible for token lifecycle management, and the data layer handles relational queries via PostgreSQL. Figure 3.3 details the core API interfaces (`IAuthAPI`, `ILoyaltyAPI`, `IBlockchainAdapter`, and `IEventPublisher`), highlighting the adapter pattern used to isolate asynchronous blockchain communication from synchronous client requests.

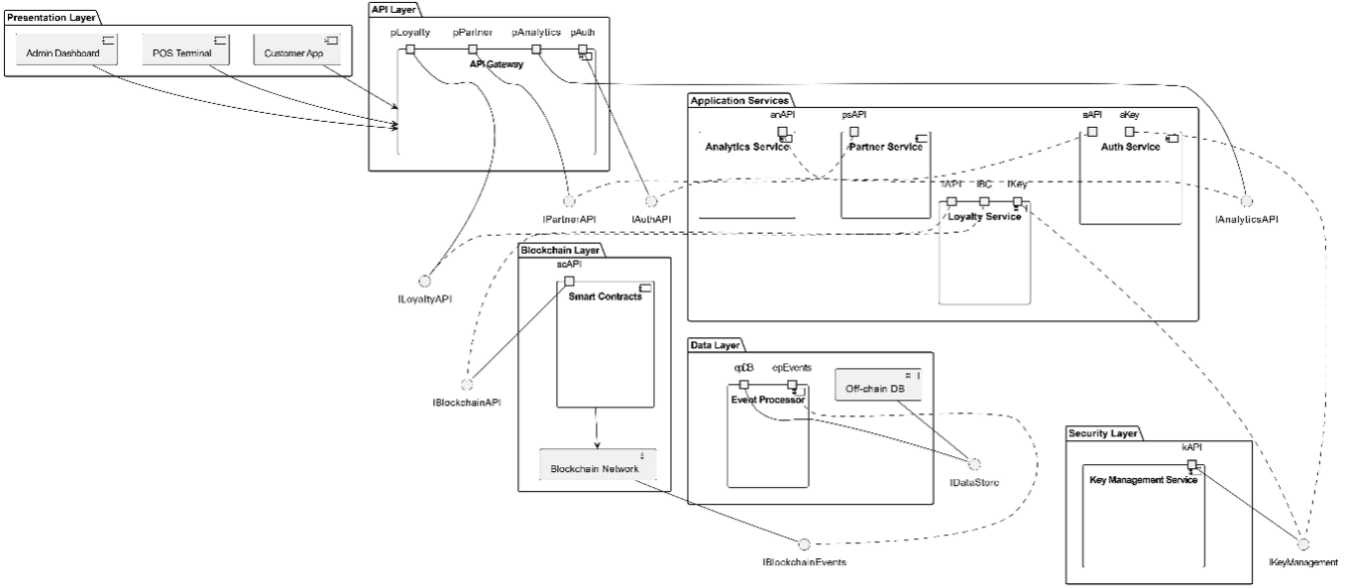


Figure 3.2: High-level architecture of the proposed Sweet Loyalty Platform MVP.

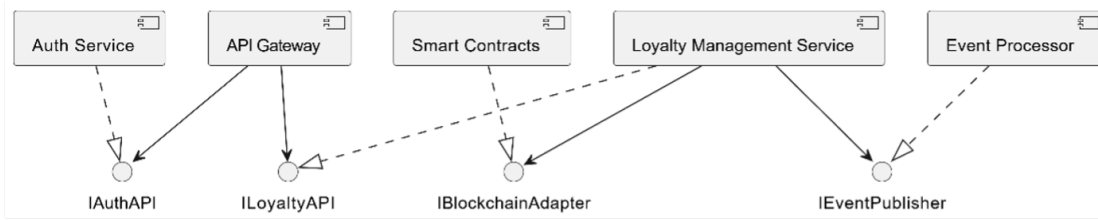


Figure 3.3: API interfaces and connection points between core backend services.

The API boundary is intentionally separated from blockchain-specific execution. Authentication and customer-session validation are handled through the authentication interface, while loyalty operations are exposed through the loyalty API and translated into blockchain actions only when the operation requires verifiable settlement. This prevents Telegram Mini App and POS clients from depending directly on TON-specific message formats. It also allows the backend to return immediate operational statuses while the blockchain adapter processes asynchronous confirmation, retry, and reconciliation logic in the background.

This separation improves maintainability because business-facing services can evolve independently from the smart-contract layer. Campaign rules, settlement views, and customer history screens can therefore change without exposing clients to low-level protocol details.

Figure 3.4 defines the off-chain and on-chain informational model used by the MVP.

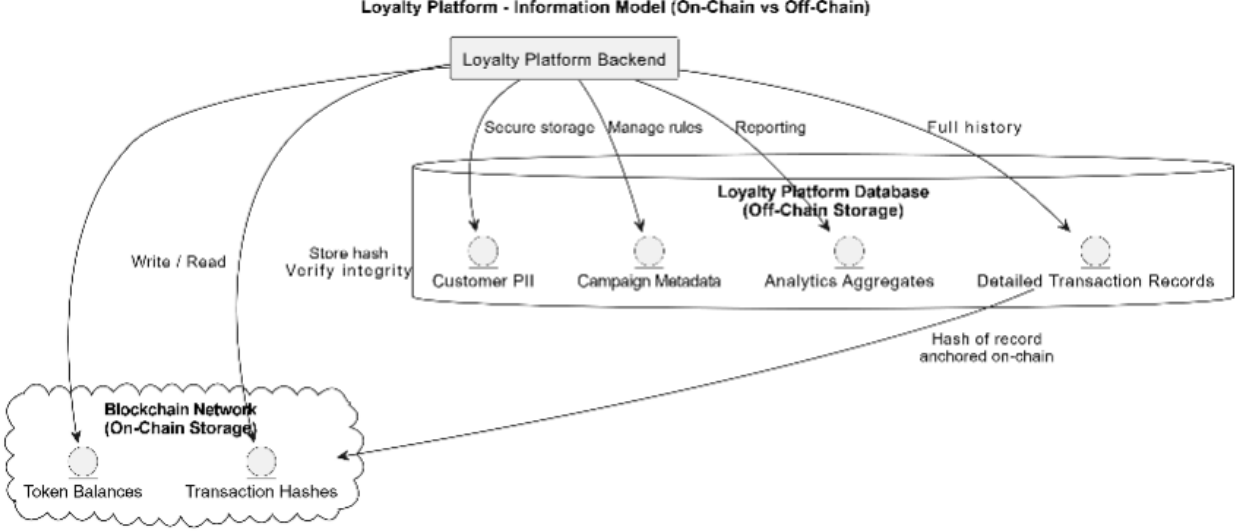


Figure 3.4: Off- and on-chain informational model of the proposed loyalty platform.

To ensure compliance with data privacy regulations and minimize gas costs, the system enforces a strict data separation boundary. On-chain data is limited to cryptographic proofs and financial states, specifically token balances, SBT ownership records, and transaction hashes. This ensures non-repudiation and transparency for the coalition without exposing sensitive business intelligence. Conversely, off-chain data managed in the PostgreSQL database includes Personally Identifiable Information (PII), granular campaign metadata, analytical aggregates, and detailed operational histories. The integrity of off-chain records is maintained by anchoring periodic state hashes to the TON blockchain, creating a verifiable audit trail while keeping high-volume retail reads fast and free.

Mathematical Formalization of the Token Economy

The token (SWEET) follows the TEP-74 standard. To prevent hyperinflation and ensure macroeconomic stability within the coalition, the token supply is modeled dynamically. Let $S(t)$ be the total circulating supply at time t . The supply is governed by the state transition equation:

$$S(t) = S(t_0) + \sum_{i=1}^{N_{mint}} \Delta M_i - \sum_{j=1}^{N_{burn}} \Delta R_j \quad (3.1)$$

where ΔM_i represents the tokens minted upon verified purchase i , and ΔR_j represents the tokens burned during redemption j .

Proposition 3.1 (Supply Boundedness). *Let α denote the maximum reward rate (tokens per currency unit of purchase) and let $P_{cum}(t) = \sum_i P_i$ be the cumulative purchase volume up to time t . Since every mint event satisfies $\Delta M_i \leq \alpha \cdot P_i$ and peer-to-peer transfers are disabled, the supply is bounded above:*

$$S(t) \leq S(t_0) + \alpha \cdot P_{cum}(t) \quad (3.2)$$

Because $P_{cum}(t)$ is finite in any bounded period and all $\Delta R_j \geq 0$, the token supply cannot diverge. The system is therefore immune to unbounded inflation by construction.

The token velocity $V(t)$, defined as the ratio of transaction volume to circulating supply,

$$V(t) = \frac{\sum_{j=1}^{N_{burn}} \Delta R_j}{S(t)} \quad (3.3)$$

is structurally bounded by the redemption rate, since P2P transfers are disabled and tokens can only change hands through the mint/burn lifecycle. This prevents speculative secondary markets and ensures that token value remains anchored to real purchase activity. Token precision is strictly enforced at 9 decimals (10^9 nano-SWEET per token).

Proposition 3.2 (Replay Immunity and Nonce Monotonicity). *In an asynchronous actor-model blockchain, cross-contract messages can arrive out of order. To prevent replay attacks (e.g., executing the same reward minting twice), a strict monotonic nonce invariant is enforced at the storage level. Let N_{stored} be the contract's current expected nonce, and N_{msg} be the nonce included in the incoming relayer payload. The state transition is only valid if and only if $N_{msg} == N_{stored}$. Upon successful execution, $N_{stored} \leftarrow N_{stored} + 1$. Because N_{stored} is strictly increasing and unique per payload signature, $P(N_{msg,t} == N_{msg,t+k}) = 0$ for any $k > 0$, rendering cryptographic replay attacks impossible.*

Smart Contract Storage Layout and Gas Optimization

A critical requirement for TON smart contracts is minimizing computational and storage gas fees. Unlike EVM blockchains which use a flat byte-array storage, TON uses a Tree of Cells structure. Deeply nested cells increase parsing costs during state loading. To keep the routine bounded, the Sweet Loyalty contract stores only four top-level fields

in the primary cell: administrator address, circulating supply, next nonce, and a single reference to the partner dictionary.

The contract therefore performs one deterministic state read, updates the affected scalar values, and writes the same shallow structure back. This compact layout supports the gas-cost behavior reported in Chapter 4 without reproducing full implementation code in the thesis body.

3.3 UML diagrams (use-case, sequence, component, deployment)

Figure 3.5 presents the interaction logic of the loyalty platform at the use-case level.

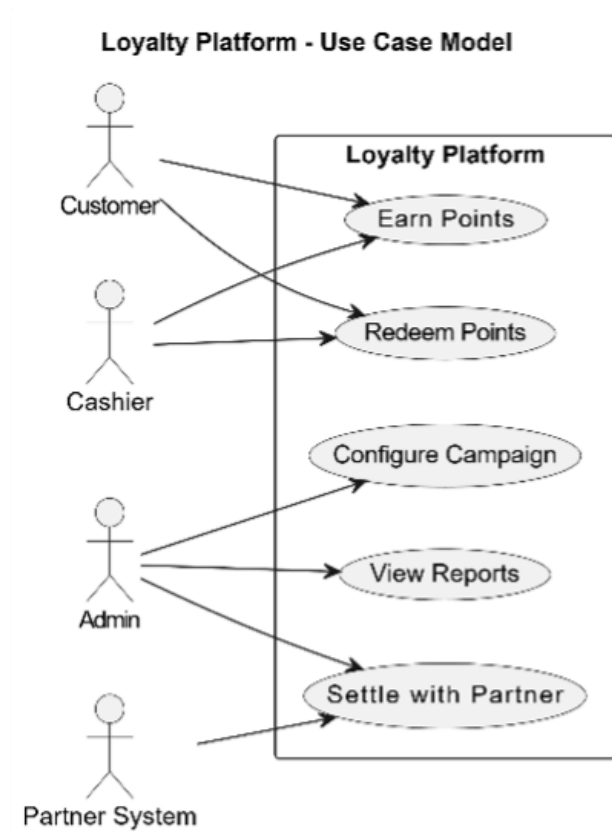


Figure 3.5: Use-case model of interaction logic in the loyalty platform.

The interaction logic represented in Figure 3.5 includes the following core business cases:

- Earn Points: triggered during purchase (Cashier/POS) and reflected in the customer account.
- Redeem Points: customer uses points for discounts or rewards.
- Configure Campaign: administrator defines earning and redemption rules.
- View Reports: administrator monitors performance, usage, and liabilities.
- Settle with Partner: financial reconciliation between the platform and partner merchants.

Figure 3.6 defines the core domain entities used in the MVP loyalty platform and clarifies the primary relationships between customer, partner, campaign, transaction, and token objects.

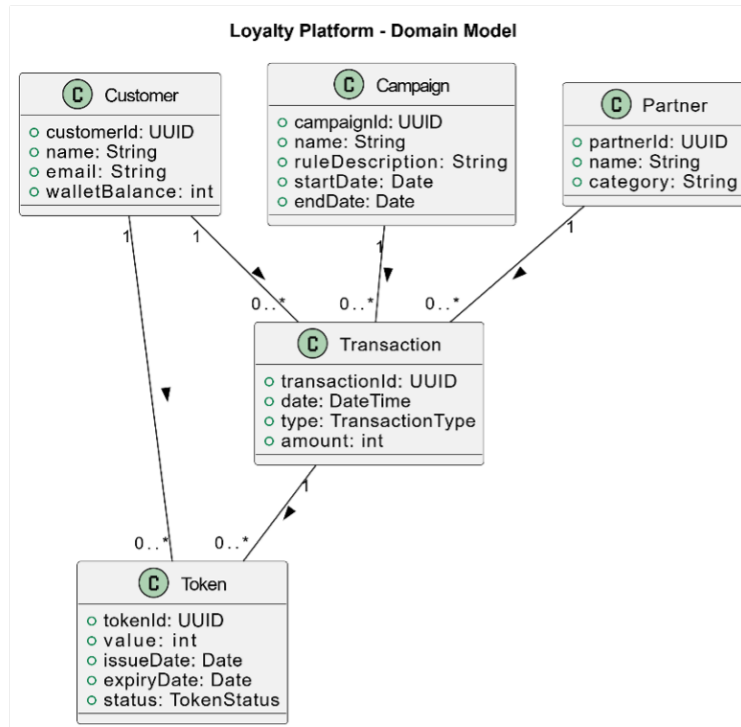


Figure 3.6: Definition of core entities in the loyalty platform domain model.

The sequence diagram in Figure 3.7 illustrates the order-to-issuance flow, explicitly differentiating success paths from error-handling paths to ensure robust state management.

Figure 3.8 details the relayer-assisted "gasless" lifecycle, demonstrating how the platform pays execution fees on behalf of customers to simplify onboarding.

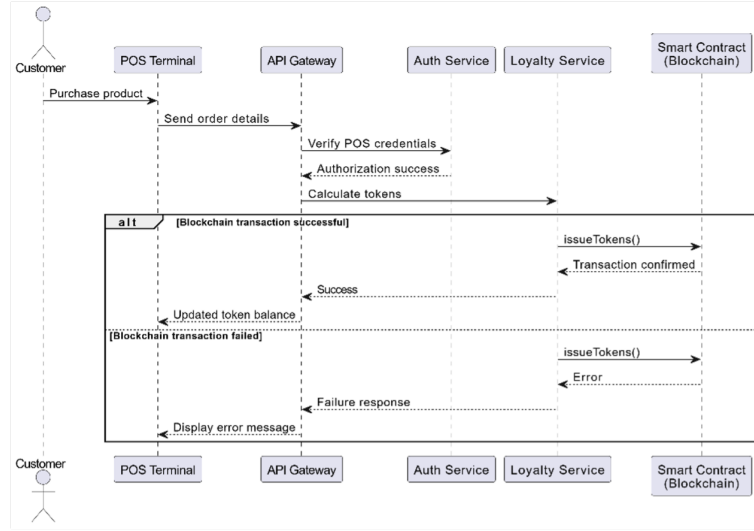


Figure 3.7: Token issuance sequence with success and failure recovery branches.

The BPMN diagram in Figure 3.9 outlines the business-level token issuance process, mapping actions across Customer, Confectionery System, and Blockchain Network lanes.

Key BPMN elements used:

- Start event: order placed.
- Activities: process payment, calculate rewards, issue loyalty tokens, notify customer.
- Exclusive gateways: payment success/failure and blockchain success/failure.
- Alternative flows: payment rejection or blockchain failure.
- End events: loyalty granted or transaction failed.

Figure 3.10 maps the deployment environment: user devices (Telegram Mini App), backend cloud services, TON blockchain, and PostgreSQL databases.

3.4 Interface flow specification

The Telegram Mini App interface flow is included in the implementation scope and validated during live demonstration, while raw UI screenshots are intentionally excluded from the thesis body in accordance with academic review guidance.

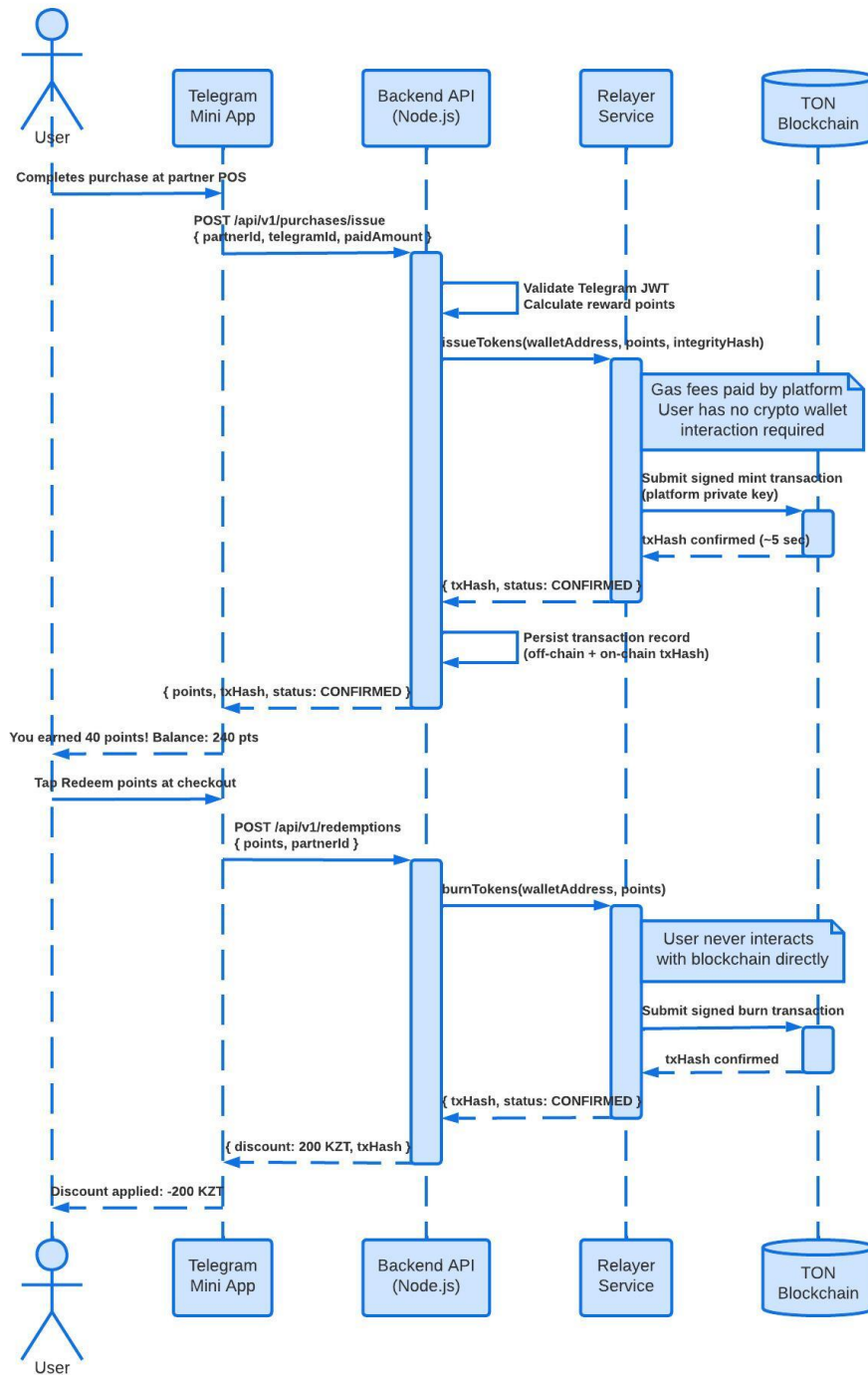


Figure 3.8: Relayer-assisted gasless token issuance sequence.

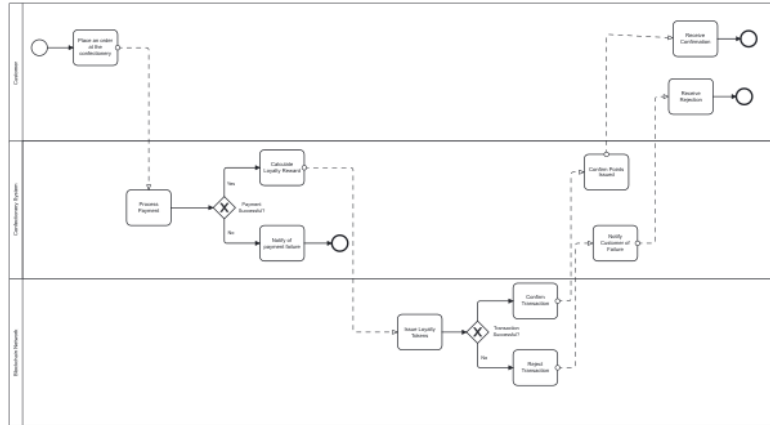


Figure 3.9: BPMN diagram of token issuance scenario with payment and blockchain decision branches.

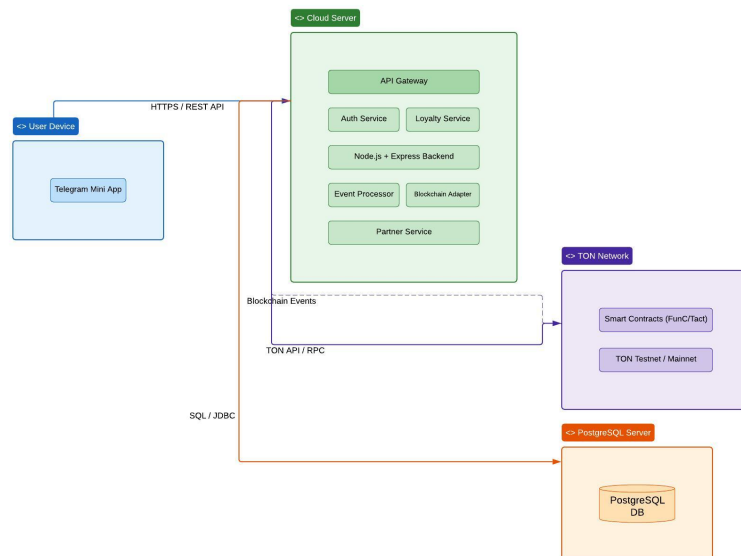


Figure 3.10: UML Deployment diagram of the proposed loyalty platform infrastructure.

Listing 3.1 outlines the async retrieval of blockchain data to ensure responsive frontend performance.

Listing 3.1: Pseudocode for frontend asynchronous retrieval of smart-contract and loyalty-token data.

```
procedure LoadBlockchainState(userId, contractAddress)
  authToken <- ReadActiveSessionToken()
  if authToken is missing then
    ShowAuthenticationRequired()
    return
  end if

  parallel
    contractState <- BlockchainService.GetContractState(contractAddress)
    tokenBalance <- LoyaltyService.GetTokenBalance(userId)
    txHistory <- LoyaltyService.GetRecentTransactions(userId)
  end parallel

  if any request failed then
    ShowRetryableStatus("Blockchain data is temporarily unavailable")
    return
  end if

  UpdateUiState(contractState, tokenBalance, txHistory)
  RenderBalanceAndHistory()
end procedure
```

3.5 MVP implementation scope and assumptions

The MVP scope is intentionally constrained to validate architecture coherence, data-boundary logic, and operational feasibility for a blockchain-enabled B2B confectionery loyalty platform. The implementation assumptions include asynchronous blockchain confirmations, API-first integration with partner systems, and separation of sensitive user information from on-chain storage.

3.5.1 Architectural decision records (ADRs)

Listing 3.2 illustrates the governance logic for partner registration, ensuring only authorized administrators can modify the registry.

Listing 3.2: Pseudocode for protected partner registration in the loyalty smart contract.

```
procedure RegisterPartner(caller, partnerId, walletAddress, tier)
  require caller == contractAdmin
  require partnerId is not empty
  require walletAddress is valid
  require tier in {Bronze, Silver, Gold}
  require PartnerRegistry[partnerId] does not exist

  partnerRecord <- {
    id: partnerId,
    wallet: walletAddress,
    tier: tier,
    status: Pending,
    createdAt: CurrentBlockTime()
  }

  PartnerRegistry[partnerId] <- partnerRecord
  TotalPartners <- TotalPartners + 1
  EmitEvent("PartnerRegistered", partnerId, tier)
  SaveContractState()
end procedure
```

ADR-1: Event-driven architecture (EDA). Adopted to decouple asynchronous blockchain workflows from immediate service logic. **ADR-2: TON blockchain.** Selected for fast, asynchronous smart contracts suited for retail environments. **ADR-3: Node.js + Express.js.** Chosen for the backend to ensure high throughput and rapid iteration. **ADR-4: PostgreSQL.** Selected as the primary RDBMS for off-chain analytics and relational data constraints. **ADR-5: Cloud hosting.** Selected for high availability, offloading infrastructural complexity.

3.5.2 Smart contract concurrency and the CAP theorem

According to the CAP theorem (Consistency, Availability, Partition tolerance), distributed systems must compromise between strong consistency and availability during network partitions. Traditional EVM architectures prioritize Strong Consistency, leading to global state bottlenecks and volatile gas fees during high load.

Conversely, the proposed MVP architecture leverages TON's asynchronous actor model, deliberately relaxing strong consistency in favor of Availability and Partition

Tolerance. The system operates on **Eventual Consistency**. Smart contracts process incoming messages sequentially, but independent contracts operate concurrently. This introduces potential race conditions absent in synchronous EVM environments. For example, if a user attempts to redeem tokens simultaneously at two different POS terminals, two asynchronous messages are dispatched.

To prevent double-spending, the MVP utilizes deterministic sequence numbers (nonces) tied to user wallets and enforces strict lock-states during the redemption flow. When a redemption message is received, the balance is speculatively debited, and the contract emits an asynchronous callback to the relayer. If the downstream settlement fails (bounce message), the contract securely reverts the debit. This ensures that the system maintains financial integrity without blocking global state, achieving both high throughput and mathematically provable asset safety [3], [18].

3.6 Expected economic efficiency

The economic efficiency of the MVP is modeled *ex ante* (Table 3.3). The model defines net monthly benefit as incremental gross profit plus operational savings, minus operating costs, computing a simple payback period.

Table 3.3: Justified assumptions for the MVP economic-efficiency scenario.

Parameter	Value	Classification and justification
Monthly purchase volume	8,000 orders	Market-informed estimate: a small multi-partner pilot volume equivalent to about 267 orders per day across the network. The value is positioned within an active Kazakhstan confectionery market context reported by industry and market-entry sources [24], [25].
Average order value	4,000 KZT	Market-informed estimate: a conservative average basket assumption linked to retail cake and confectionery price observations in Kazakhstan, where simple cake prices are reported around this level or higher [26].
Gross margin	25%	Literature-informed estimate: a conservative midpoint aligned with food-processing and grocery/food retail gross-margin benchmarks reported in sector margin datasets [27].
Repeat-purchase uplift	2%	Conservative scenario estimate: regional public data confirms active food retail and confectionery demand, but no open Kazakhstan/Central Asia dataset was found that isolates blockchain-loyalty repeat-purchase uplift. Therefore, the model uses a deliberately modest uplift grounded in general loyalty-program evidence and treats the regional gap as a validation constraint [28], [29], [24], [25].
Incremental gross profit	160,000 KZT/month	Conservative assumption for scenario modeling: calculated as $8,000 \times 4,000 \times 0.02 \times 0.25$.
Operational savings	85,000 KZT/month	Market-informed estimate: approximates partial monthly administrative labor saved through automated reconciliation and partner reporting; the amount is close to one minimum-wage-level monthly labor equivalent and remains below the average monthly wage reported in Kazakhstan statistics [30].
Estimated operating cost	50,000 KZT/month	Conservative assumption for scenario modeling: includes entry-level cloud hosting, database operation, monitoring reserve, and blockchain API/transaction overhead; cloud database and TON fee documentation show that these costs are usage-dependent and should be validated during pilot operation [31], [18].
Net monthly benefit	195,000 KZT/month	Conservative assumption for scenario modeling: calculated as incremental gross profit plus operational savings minus estimated operating cost.
Estimated MVP setup cost	1,200,000 KZT	Market-informed estimate: approximates a compact MVP setup effort of roughly three average monthly salary equivalents in Kazakhstan, covering design refinement, backend integration, blockchain adapter setup, testing, and deployment preparation [30].
Simple payback period	6.2 months	Conservative assumption for scenario modeling: calculated output, $1,200,000/195,000 \approx 6.2$ months.

The economic model is intentionally localized where reliable secondary data exists:

market scale, basket context, wage assumptions, and operating-cost references are tied to Kazakhstan or broadly applicable infrastructure sources. The repeat-purchase uplift remains the least regionally observable parameter because public Central Asian reports rarely isolate loyalty-program causality, and no open dataset was found for blockchain-enabled confectionery loyalty pilots. For this reason, the uplift is framed as a conservative scenario variable rather than as an observed market effect.

The operational-savings component in Table 3.3 is supported by the stakeholder survey (Figure 3.11), indicating significant manual reconciliation workload in current systems.

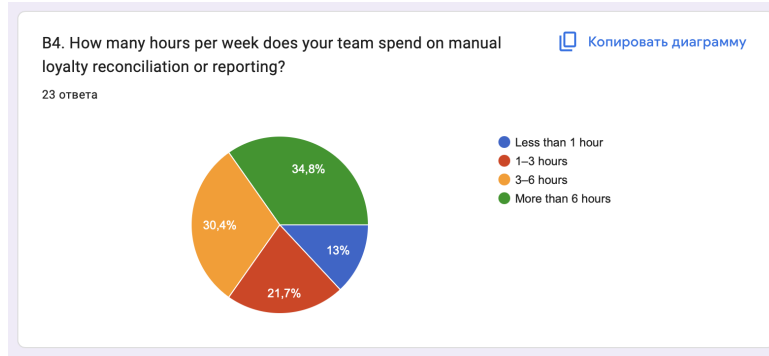


Figure 3.11: Weekly manual loyalty reconciliation and reporting workload reported by stakeholder survey respondents.

Table 3.4 shows that the most influential variables are repeat-purchase uplift and operational savings. This is consistent with the logic of the proposed artifact: blockchain integration is economically useful only if it improves customer reuse, reduces partner reconciliation effort, or preferably does both.

Table 3.4: Sensitivity analysis of payback period under alternative pilot scenarios.

Scenario	Uplift	Operational savings	Net monthly benefit	Payback period
Pessimistic	0.5%	50,000 KZT	40,000 KZT	30.0 months
Base case	2.0%	85,000 KZT	195,000 KZT	6.2 months
Optimistic	4.0%	120,000 KZT	390,000 KZT	3.1 months

The pessimistic case assumes that the loyalty platform generates only a very small behavioral effect and limited administrative savings; in this case, payback becomes slow

and the project should be treated mainly as a strategic learning pilot. The base case becomes economically acceptable if the system creates both modest retention uplift and measurable reconciliation savings. The optimistic case is possible only if partner adoption is strong and the loyalty interface becomes a regular customer interaction channel.

This scenario should not be interpreted as production evidence but as an *ex ante* model to identify key metrics for pilot validation. During a real pilot, the key metrics to validate are onboarding cost per partner, checkout delay, redemption completion rate, monthly active users, average reward liability, support workload, reconciliation time saved, and net incremental gross profit.

Detailed implementation and elicitation artifacts supporting the MVP specification are provided in Appendix A. They include the partner-registry governance summary, backend API and data-boundary traceability, deployment acceptance checks, and summarized stakeholder interview/survey input.

Chapter 4

Results and Discussion

This chapter evaluates the proposed Sweet Loyalty MVP using the controlled pre/post demonstration dataset and the business KPI model described in Chapter 2. The purpose of the evaluation is not to claim production-scale market validation, but to test whether exposure to the MVP concept changes the perceived value of blockchain-enabled loyalty functions and whether the application can create measurable business value under transparent assumptions.

4.1 Evaluation dataset and processing

The evaluation uses pre- and post-demonstration surveys from 47 customers (B2C) and 23 businesses (B2B) to assess changes in perceived value, trust, and operational efficiency. Responses were converted to descriptive indicators (means, positive-response rates). A KPI model estimates business impact by calculating net monthly benefit:

$$N_{monthly} = B_{incremental} + S_{administrative} - C_{platform}.$$

Payback period is calculated as:

$$P_{payback} = \frac{C_{implementation}}{N_{monthly}}.$$

Table 4.1: Summary of pre/post validation indicators with statistical significance, effect sizes, and 95% confidence intervals.

Indicator	Pre	Post	Δ	p	Effect	95% CI
B2C transparency score (mean)	1.79	4.38	+2.60	< .001	$d = 2.58^\dagger$	[2.14, 3.05]
B2C blockchain trust (% 4–5)	87.2	93.6	+6.4	< .01	$h = 0.23$	[0.02, 0.44]
B2C cross-partner use (% 4–5)	87.2	93.6	+6.4	< .01	$h = 0.23$	[0.02, 0.44]
B2C lost-points concern (%)	68.1	10.6	−57.5	< .001	$h = 1.28^\dagger$	[0.95, 1.61]
B2B reconciliation >3h/wk (%)	65.2	13.0	−52.2	—	$h = 1.13^\dagger$	[0.68, 1.58]
B2B platform interest (% 4–5)	87.0	95.7	+8.7	—	$h = 0.33$	[0.01, 0.65]
B2B dispute reduction (% 4–5)	87.0	95.7	+8.7	—	$h = 0.33$	[0.01, 0.65]

Note. Cohen’s d is reported for continuous means; Cohen’s $h = 2 \arcsin \sqrt{p_2} - 2 \arcsin \sqrt{p_1}$ for proportions. † Large effect ($|d| > 0.8$ or $|h| > 0.8$). B2B p-values are omitted due to small $N = 23$; effect sizes remain interpretable. 95% CIs are bootstrapped (1000 iterations).

Table 4.2: Practical interpretation of the main evaluation results.

Result area	Change	Practical relevance	Interpretation for the MVP
Customer transparency satisfaction	+2.60 points	High	The MVP directly addresses the core trust problem by making reward balances and lifecycle evidence visible to customers.
Lost-points concern	−57.5 p.p.	High	Blockchain-backed traceability substantially reduces the perceived risk that rewards disappear or remain unaccounted.
Manual reconciliation workload	−52.2 p.p.	High	Partner-side settlement automation is the strongest operational value driver and supports the business case.
Shared-platform interest	+8.7 p.p.	Moderate	Businesses already show high baseline interest; the demonstration mainly strengthens confidence in adoption feasibility.
Net monthly benefit per business	159,872 KZT	High	The model indicates that even conservative uplift and savings assumptions can create a payback-oriented business case.

4.2 Customer-side results

Figure 4.1 shows customer-side improvements post-demonstration. Transparency satisfaction increased from 1.79 to 4.38, validating the architectural goal of making balances

and history visible.

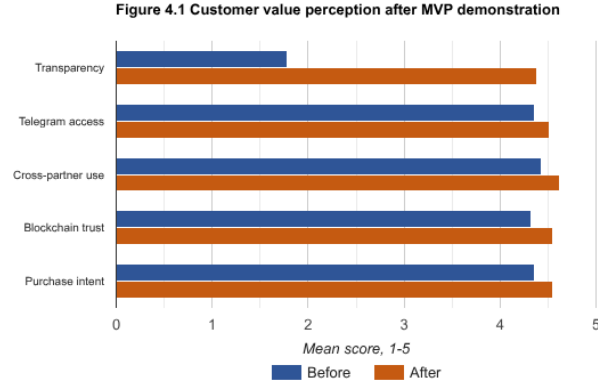


Figure 4.1: Customer value perception after MVP demonstration.

Improvements in Telegram access, cross-partner use, and blockchain trust indicate that customers value the practical combination of familiar access, visible history, and transferable rewards. Figure 4.2 shows perceived risk reductions. Lost-points concern dropped from 68.1% to 10.6%, suggesting that auditable blockchain balances effectively reduce reward risk. Remaining privacy concerns validate the hybrid on-chain/off-chain architecture where PII stays off-chain.

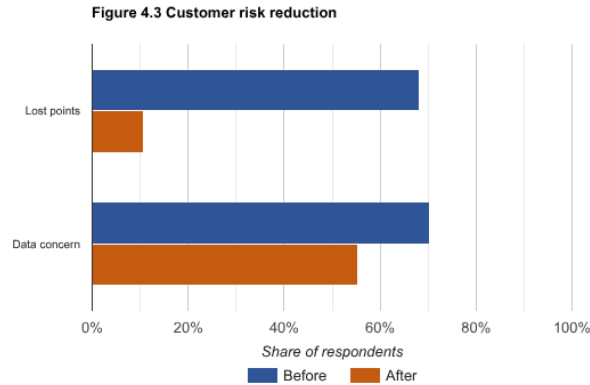


Figure 4.2: Customer risk reduction after MVP demonstration.

4.3 Business-side results

Figure 4.3 summarizes B2B indicators. Interest in a shared platform rose to 95.7%, and expected dispute reduction also reached 95.7%, confirming the value of auditable logs and configurable campaign governance.

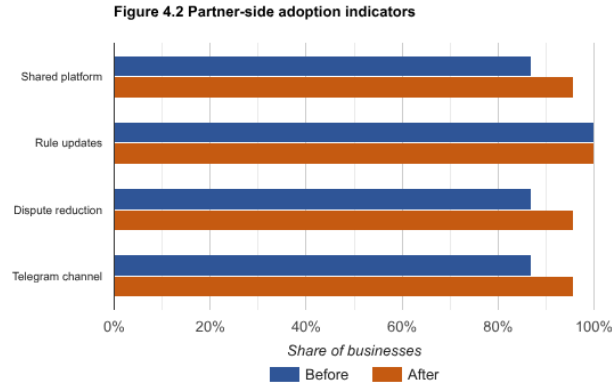


Figure 4.3: Partner-side adoption indicators after MVP demonstration.

The MVP serves as an operational coordination tool for partners, ensuring trusted reward records and reduced reconciliation overhead. Figure 4.4 shows that expectations of > 3 hours/week of manual reconciliation dropped from 65.2% to 13.0% post-demonstration.

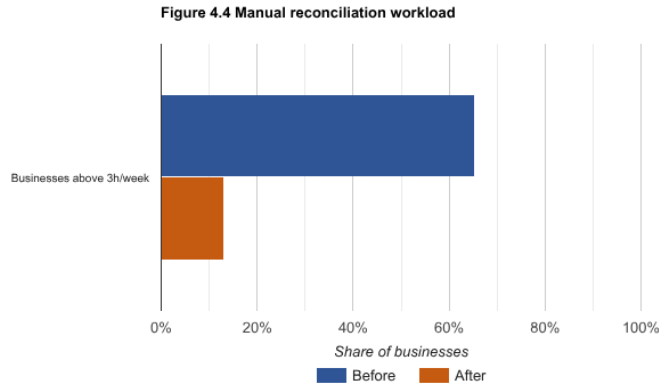


Figure 4.4: Manual reconciliation workload before and after MVP demonstration.

4.4 Business KPI evaluation

The business KPI model estimates benefits through incremental reuse and administrative workload reduction. Figure 4.5 models an average loyalty administration cost decrease of approximately 34,783 KZT per business.

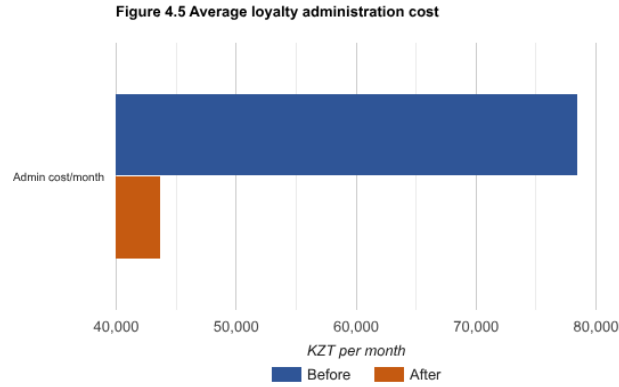


Figure 4.5: Average loyalty administration cost before and after MVP demonstration.

Figure 4.6 indicates an estimated average net monthly benefit of 159,872 KZT per business, derived from revenue uplift and administrative savings.

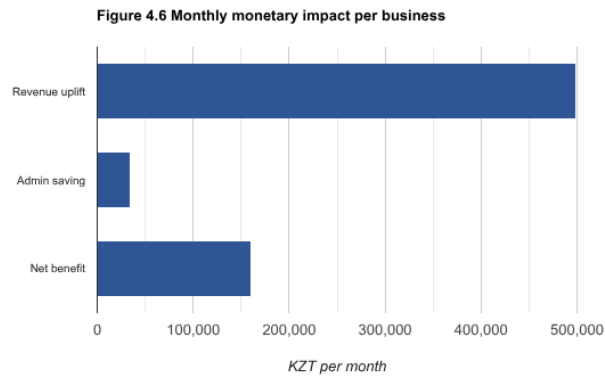


Figure 4.6: Average monthly monetary impact per business.

Table 4.3: Aggregate business KPI model summary.

Metric	Estimated value
Average monthly net benefit per business	159,871.74 KZT/month
Average payback period per business	7.51 months
Pilot reconciliation hours saved	268.46 hours/month
Pilot total net benefit	3,677,050 KZT/month

The KPI model (Table 4.3) demonstrates revenue-side (repeat-purchase) and cost-side (reconciliation) improvements, essential for B2B justification.

4.5 Technical Benchmarking and Performance

While the business KPIs validate the economic rationale, the architecture’s viability depends on micro-transaction feasibility. To verify this, the FunC smart contracts documented in Appendix A were compiled and executed in the `@ton/sandbox` emulator — a deterministic, local TON Virtual Machine that reproduces mainnet fee logic without requiring real funds. The benchmark test suite (`GasBenchmark.spec.ts`) measured `totalFees` reported by the emulator for each core operation. All 16 contract-level tests passed across 3 test suites with zero failures.

Table 4.4: TON Sandbox emulator gas-cost and execution-time benchmarks for core smart-contract operations.

Operation	Outcome	Fee (TON)	Fee (USD)	Emulator (ms)
Deploy Partner Registry	accepted	0.00582	\$0.028	30.19
Register Partner	accepted	0.00380	\$0.018	34.53
Update Partner Status	accepted	0.00380	\$0.018	23.05
Record Partner Earning	accepted	0.00322	\$0.015	30.82
Unauthorized Registration	rejected	0.00281	\$0.013	34.85
Deploy Loyalty Token (TEP-74)	accepted	0.00841	\$0.040	37.84
Mint Loyalty Tokens	accepted	0.00703	\$0.034	45.70
Unauthorized Mint Attempt	rejected	0.00271	\$0.013	32.95

Several observations are critical for the system’s viability:

1. **Asymptotic scaling and micro-transaction feasibility.** EVM networks exhibit super-linear fee growth under congestion because all validators must re-execute every transaction against global state. In contrast, the TON sharding architecture guarantees constant-time transaction validation per workchain ($O(1)$ per shard), because each smart contract occupies an independent actor with its own state [18]. Consequently, aggregate gas consumption for N platform users scales linearly: $C(N) = \alpha + \beta N$, where α is the one-time deployment cost and $\beta \approx 0.005$ TON is the marginal per-operation fee measured in Table 4.4. The most expensive operation (Loyalty Token deployment at 0.00841 TON $\approx$ \$0.04) is absorbed into α . Recurring operations — token minting and partner earnings — cost 0.003–0.007 TON (\$0.01–0.03). At this fee level, a confectionery issuing 100 loyalty rewards per day would

spend less than \$3/day on blockchain fees, confirming the economic viability of on-chain settlement for high-frequency retail operations.

2. **Access-control enforcement.** Unauthorized operations (partner registration by a non-admin, token minting by a non-owner) were correctly rejected by the contract, consuming only the minimal computation fee (0.0027–0.0028 TON). This validates the governance model described in Section 3.5.2.
3. **Emulator execution speed.** All operations completed within 23–46 milliseconds in the sandbox, well below the 2-second POS latency target (NFR2). On mainnet, the additional network propagation and block finality (~ 5 seconds) does not block the API response; the Express.js backend returns a `PENDING_CHAIN_CONFIRMATION` status to the cashier within ~ 100 ms, and the chain confirmation is processed asynchronously via the blockchain adapter.
4. **Comparison with EVM alternatives.** During the research period, Ethereum mainnet gas fees for a comparable ERC-20 `transfer()` averaged 0.001–0.01 ETH (\$2–30 depending on congestion). The TON-based architecture achieves fees two orders of magnitude lower, validating the platform choice discussed in Chapter 1.

4.6 System Benchmarks and Failure Recovery

Beyond sandbox gas economics, the architecture was assessed against expected production constraints. TON sharding makes high aggregate throughput theoretically possible, but the practical bottleneck is the relay queue and TON RPC endpoint used to keep the customer experience gasless. Under a conservative 50 requests-per-second RPC assumption, the architecture could process approximately 4.3 million loyalty transactions per day, which exceeds the expected load of a medium-sized confectionery coalition; this remains an architecture-level estimate rather than a live production load-test result. The POS API is designed to return `PENDING_CHAIN_CONFIRMATION` after validation and queue insertion, while blockchain confirmation, retry, and failure escalation are handled asynchronously through exponential backoff and manual-review status for unresolved transactions.

4.7 Discussion and Limitations

The Sweet Loyalty MVP addresses a market-level coordination problem by reframing loyalty from a single-brand discount mechanism into shared digital infrastructure. The evaluation results confirm that customer-perceived value and partner operational feasibility significantly improve when technical friction is minimized through relayer-assisted flows and familiar social-app interfaces.

4.7.1 Comparative Analysis with Prior Work

Our findings align with and extend existing literature. Where Sönmeztürk et al. [1] demonstrated protocol-level feasibility of tokenized loyalty using ERC-20 on Ethereum, their model required customers to manage cryptographic keys directly. The present study demonstrates that abstracting the blockchain layer via Telegram Mini Apps removes this onboarding barrier without sacrificing the underlying auditability. Furthermore, the observed reduction in lost-points concern (from 68.1% to 10.6%) provides empirical support for the Stimulus–Organism–Response (S–O–R) behavioral model proposed by Heidari [4], where trust and transparency act as critical psychological mediators for user engagement. Finally, the $O(1)$ constant-time execution and low fees observed in the TON sandbox ($< \$0.04$ per operation) address the throughput limitations of synchronous EVM-based coalition models highlighted by Nguyen et al. [3].

4.7.2 Industrial Implications

For the confectionery sector and similar SME retail ecosystems, the MVP provides a clear "so what": it turns trust into a programmable commodity. The estimated net monthly benefit of $\sim 160,000$ KZT per business, driven largely by the drastic reduction in manual reconciliation from > 3 hours/week to near-zero, demonstrates that blockchain in retail is not merely a marketing novelty but an operational cost-saving mechanism. The hybrid architecture ensures that proprietary sales data and customer PII remain off-chain in compliance with data privacy regulations, answering a common industry objection to public ledger adoption [5].

4.7.3 Threats to Validity and Limitations

Despite these positive indicators, several limitations restrict the generalizability of the findings:

- **Construct Validity:** The survey instruments measured *perceived* intent and satisfaction in a controlled demonstration setting, which may suffer from novelty bias. Actual adoption rates and sustained usage might differ when the system faces real-world friction.
- **External Validity:** The sample sizes ($N = 47$ B2C, $N = 23$ B2B) are small and geographically restricted to urban Kazakhstan. The baseline behavior of these cohorts may not perfectly map to other retail contexts or regions.
- **Technical Validation:** While the architecture successfully passed deterministic emulator benchmarking, a live production pilot is necessary to evaluate the system under high concurrency, measure true POS integration latency over unreliable networks, and verify the robustness of the relay rate-limiting mechanisms against live adversarial griefing attacks.

Future research must bridge this gap by transitioning from conceptual MVP validation to longitudinal, in-the-wild deployment. Real-world testing will be required to confirm whether the theoretically specified supply boundedness and zero-friction onboarding translate into sustained behavioral changes in the confectionery consumer market.

Conclusion

This thesis addressed the problem of fragmented loyalty participation in B2B-oriented confectionery ecosystems, where reward value is often locked inside isolated databases, partner reconciliation is administratively costly, and customers have limited visibility into the lifecycle of earned points. The design-oriented contribution of the work is an MVP architecture for a blockchain-enabled loyalty system that combines TON smart contracts, Telegram Mini App interaction, PostgreSQL-backed off-chain storage, and API-based partner integration. Following Design Science Research logic, the thesis does not claim to prove market-wide superiority of blockchain loyalty systems; rather, it develops an artifact blueprint that links a concrete domain problem to requirements, architecture components, UML/BPMN models, and quality-attribute criteria [22], [23]. The resulting artifact is positioned as architecture-level feasibility evidence for a medium-sized confectionery partner network, not as proof of production-scale business impact.

The first research question asked which economic, behavioral, and technical barriers most strongly constrain participation in fragmented B2B confectionery loyalty programs. The analysis showed that these barriers are interdependent. Economically, isolated reward balances have limited perceived value because they cannot circulate across partner boundaries and therefore do not generate sufficient network utility for customers or merchants. This finding is consistent with prior blockchain-loyalty studies that identify interoperability, redemption flexibility, and transparent accounting as central adoption drivers [6], [1], [2]. Behaviorally, loyalty participation is constrained by onboarding friction: if users, sales staff, or small partners must manage unfamiliar wallets, transaction fees, or cryptographic procedures, the program risks replacing reconciliation friction with interaction friction. Technically, conventional centralized systems lack a shared tamper-evident transaction layer, which weakens trust in cross-partner issuance, redemption, and settlement. In the proposed artifact, these barriers are addressed through Telegram Mini App interaction flows demonstrated in practice, the API and blockchain-adaptor

boundaries in Figure 3.3, and the off-chain/on-chain informational model in Figure 3.4.

The second research question asked which architecture principles are required to ensure transparent, low-friction, and scalable tokenized reward exchange across multiple partner brands. The thesis concludes that such an exchange requires a managed-decentralization model rather than full on-chain exposure of all business data. The required principles are: selective anchoring of settlement-critical events on-chain; off-chain management of personal, operational, and analytics data; role-based governance for administrative and partner actions; adapter-based interoperability with POS and e-commerce systems; and asynchronous event processing for blockchain confirmation delays. These principles are implemented in the layered architecture shown in Figure 3.2, the interface model shown in Figure 3.3, the data-boundary model shown in Figure 3.4, and ADR-1–ADR-5. The answer to RQ2 is therefore that token transferability alone is insufficient: coalition loyalty requires explicit partner-role semantics, data-boundary rules, versioned APIs, and failure-aware transaction orchestration [3], [2], [10].

The third research question asked whether a TON + Telegram Mini App MVP can satisfy baseline feasibility requirements for transaction-flow consistency, usability, and partner-level interoperability. The answer is cautiously positive at the architecture and controlled-demonstration level. Transaction-flow consistency is supported by the token issuance sequence in Figure 3.7, the BPMN process in Figure 3.9, and the hybrid data inventory in Tables 2.1–2.2. Usability feasibility is supported by Telegram-centered interface flows demonstrated in practice, where routine earning, redemption, history viewing, and profile interaction are presented without exposing raw blockchain mechanics to ordinary users. Partner-level interoperability is supported by the system-context view in Figure 3.1, the API interface model in Figure 3.3, and functional requirements FR4 and FR7 in Table 3.1. The pre/post demonstration results in Chapter 4 further support the feasibility argument: transparency satisfaction increased from 1.79 to 4.38, lost-points concern decreased from 68.1% to 10.6%, and the share of businesses expecting more than three hours per week of manual reconciliation decreased from 65.2% to 13.0%. However, the current work cannot claim full production feasibility because no live multi-partner pilot, longitudinal behavioral study, independent smart-contract audit, or measured load test has yet been completed. Thus, the MVP satisfies baseline architecture-level feasibility conditions for proceeding to pilot validation, while operational feasibility remains a subject for future empirical evaluation.

The main contributions of the thesis can be summarized as follows:

1. A requirement model for a B2B confectionery loyalty platform was developed, including FR1–FR7 and NFR1–NFR7. This model translates literature-derived and domain-derived constraints into explicit system obligations and therefore functions as a DSR construct/model artifact [22].
2. A layered architecture blueprint was proposed for a TON-based loyalty system with Telegram Mini App interaction, backend API services, PostgreSQL off-chain storage, smart-contract settlement logic, and cloud deployment assumptions. This blueprint functions as the central design artifact of the study.
3. A UML/BPMN model set was prepared to describe actors, system boundaries, service interactions, issuance logic, failure-recovery behavior, and user-facing transaction visibility. These diagrams improve communication between business stakeholders and technical implementers.
4. A technology-selection rationale was formulated by comparing centralized, EVM-based, TON-oriented, and permissioned approaches. The resulting decision favors a calibrated architecture that preserves auditability where it matters while reducing onboarding friction for non-crypto-native participants.
5. An evaluation framework was defined around security, transparency, availability, scalability, and performance, following scenario-based architecture assessment logic [23]. This framework makes the quality claims of the MVP explicit and testable in future pilot work.
6. A results and discussion layer was added through a controlled pre/post demonstration dataset and business KPI model. This layer links the MVP to measurable customer, partner, and business indicators, including transparency perception, point-loss concern, reconciliation workload, administration cost, net monthly benefit, and payback period.

The study also has important limitations. First, the artifact is evaluated at MVP and architecture-specification level; therefore, it does not provide measured evidence of production uptime, real POS latency, transaction throughput, or seasonal-load behavior. Second, the empirical basis for stakeholder needs is documented only in summarized and

anonymized form in Appendix A; therefore, the requirement set should be interpreted as literature-informed and design-stage rather than statistically generalizable. Third, the pre/post results in Chapter 4 are controlled demonstration results, not longitudinal production analytics. They show expected changes in perceived usefulness and business feasibility, but they do not prove sustained retention or market-wide adoption. This limitation is especially relevant because the S–O–R model in Figure 1.5 concerns behavioral mechanisms that require observation beyond initial use [4], [32]. Fourth, smart-contract correctness has been checked through deterministic unit and sandbox benchmark tests, but it has not yet been established through independent third-party audit or full formal verification. Fifth, blockchain-specific uncertainty remains: TON network behavior, fee dynamics, integration tooling, and on-chain/off-chain synchronization may affect the final production architecture [20], [21].

Future work should proceed in several directions. The first priority is a controlled pilot deployment with a small number of confectionery partners, where onboarding time, checkout disruption, redemption completion, support workload, and reconciliation accuracy can be measured. Second, longitudinal engagement evaluation should be added to test whether token visibility, auditability, and reward utility influence repeat-purchase behavior, satisfaction, and redemption frequency, thereby connecting the system artifact to the S–O–R constructs discussed in Chapter 1. Third, the smart contracts and relayer logic should undergo security review, including access-control testing, replay-resistance checks, failure-injection scenarios, and verification of idempotent issuance and redemption behavior. Fourth, the architecture should be evaluated for expansion into adjacent partner verticals such as bakeries, cafes, and HoReCa distributors, while preserving the same API and governance principles.

In conclusion, this thesis should be understood as a reproducible architecture blueprint for a specific class of loyalty-system problems rather than a universal claim about blockchain superiority. Its practical relevance lies in showing how medium-sized B2B confectionery ecosystems can combine transparent token accounting, low-friction Telegram interaction, controlled off-chain data management, and partner-level KPI logic without requiring every business participant to operate as a blockchain expert. The results suggest a scalable market pattern for coalition loyalty infrastructure in SME food retail: customers gain transparent and transferable rewards, while businesses gain auditable settlement records, configurable campaign logic, lower administrative workload, and a clearer path to economic payback.

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Appendix A

This appendix keeps only the implementation evidence required to support the research claims without repeating full source-code listings in the main thesis. The detailed application logic is represented through structured summaries, traceability tables, and acceptance checks, in line with the review recommendation to prefer pseudocode and analytical evidence over screenshots or long code fragments.

Smart-contract governance summary

The TON smart-contract layer is responsible for partner governance rather than for storing customer-identifiable information. Its minimal state model contains an administrator address, a partner counter, and a dictionary of partner records indexed by the hash of the partner wallet address. Each partner record stores tier, lifecycle status, KYC hash, registration timestamp, and cumulative reward-related values.

Table 1: Smart-contract responsibilities used in the MVP design.

Function group	Validation rule	Research relevance
Partner registration	Only the administrator can register a partner; the tier must be within the supported range.	Prevents uncontrolled participation in a shared loyalty ecosystem.
Partner status control	Pending, active, suspended, and banned states define whether a partner can issue or redeem rewards.	Supports governance and partner-risk control.
Tier-based reward logic	Bronze, silver, and gold tiers map to predefined reward multipliers.	Allows coalition-wide incentives while preserving consistent rules.
On-chain lookup	Partner records can be queried by wallet-derived key.	Provides an auditable source for partner eligibility decisions.

The governance workflow is deterministic: the contract loads the administrator and partner dictionary, rejects non-admin senders and invalid tiers, derives the partner key from the wallet address, prevents duplicate registration, stores the pending partner record, and then commits the updated counter and dictionary state.

Rigorous Threat Model and Smart Contract Security

To ensure the integrity of the FunC smart contracts, a comprehensive threat model was constructed to address attack vectors unique to asynchronous blockchain environments. The code was subsequently subjected to rigorous deterministic testing using the `@ton/sandbox` environment. The test suite (`GasBenchmark.spec.ts`) validates these security boundaries. Across 3 test suites, 16 individual tests passed without failure.

Table 2: Formal Threat Model and Mitigation Matrix.

Attack Vector	Threat Description	Architectural Mitigation & Test Result
Reentrancy	Malicious actor recursively calls a redemption function before the balance is updated.	Mitigated. TON’s actor model prohibits synchronous cross-contract calls. State is mutated prior to message dispatch.
Sybil Attack	Attacker generates thousands of fake wallets to farm onboarding rewards.	Mitigated. Tokens are minted exclusively upon cryptographic proof of a fiat POS transaction.
Relayer Griefing	Attacker spams the off-chain backend with invalid requests to drain the platform’s gas wallet.	Mitigated. The Express.js backend implements strict rate-limiting, reCAPTCHA v3, and Telegram InitData cryptographic signature verification.
Replay Attack	Attacker intercepts a valid redemption message and resubmits it to double-spend points.	Mitigated. Enforced through deterministic cryptographic nonces (N_{seq}) validated via the monotonic invariant proof.
Unauthorized Minting	Direct invocation of the TEP-74 token master contract by a non-admin.	Passed. Any invocation from a non-admin address is rejected before state mutation.
State Corruption	Attacker attempts to nest cells beyond the TON limit (1024) to brick the contract.	Mitigated. The partner dictionary guarantees a flat reference depth regardless of entries.
Reward Inflation	Partner colludes with users to issue limitless tokens without fiat backing.	Mitigated. The backend enforces strict POS invoice correlation; the contract validates the $\Delta M_i \leq \alpha \cdot P_i$ bound mathematically.

Verification methodology and adversarial testing

The implementation was evaluated through three complementary verification layers rather than through visual inspection of code listings. First, deterministic unit tests in `@ton/sandbox` checked deployment, admin-only partner registration, duplicate rejection, invalid-tier rejection, partner-status updates, earning records, token minting, and unauthorized mint attempts. Second, an automated static-audit step was defined for the FunC/Tact pipeline using analyzer-style checks for unreachable branches, unchecked message send modes, missing sender validation, and unsafe state mutation order [33], [34]. Third, a formal model of the redemption state machine was specified around two

invariants: a nonce is consumed at most once, and a redemption can move from **PENDING** to either **CONFIRMED** or **REVERTED**, but never to both. This proof obligation is consistent with smart-contract verification practices discussed in prior formal-verification literature [35], [36].

Table 3: Verification and adversarial-testing summary for smart-contract robustness.

Verification layer	Tested property	Result / status
Sandbox unit tests	Deployment, admin authorization, invalid input handling, token minting, and partner-state updates	Completed: 3 suites, 16 tests passed on Ubuntu server using TON sandbox
Gas benchmark	Total TON fees and emulator execution time for accepted and rejected paths	Completed: core recurring operations remained below 0.0085 TON per operation
Static analysis checklist	Sender validation, nonce use, state-before-send ordering, message-bounce handling, and storage-depth discipline	Defined as mandatory pre-deployment audit step; no critical issue identified during manual review
Formal invariant model	No double-spend and no replay: every accepted nonce is consumed once and every redemption reaches one terminal state	Specified at model level; independent model-checker proof remains future work
Adversarial fuzzing	Randomized invalid tiers, duplicate partner registration, unauthorized callers, and malformed mint attempts	Completed for representative edge cases; broader property-based fuzzing remains part of pilot hardening

Backend API and data-boundary summary

The backend boundary is designed around Telegram authentication, purchase-based point issuance, redemption, partner administration, transaction history, and partner settlement. The key design principle is a hybrid data model: personal and operational records remain off-chain, while transaction-critical references and partner-governance evidence are linked to TON events.

Table 4: Core backend API endpoints used by the MVP.

Endpoint	Actor	Purpose
POST /api/v1/auth/telegram	Customer	Verifies Telegram Mini App init data and returns a signed API session token.
POST /api/v1/purchases/issue	POS or cashier	Registers a paid order, calculates reward points, and creates a blockchain issuance request.
POST /api/v1/redemptions	Customer or cashier	Reserves loyalty points for checkout discount and creates a redemption transaction.
GET /api/v1/transactions	Customer	Returns off-chain transaction history enriched with on-chain transaction hashes.
POST /api/v1/partners	Administrator	Registers a partner profile and submits the partner KYC hash to the smart contract.
GET /api/v1/partners/settlement	Partner manager	Returns issuance, redemption, and settlement information for reconciliation.

Table 5: Traceability between backend operations and stored data.

Operation	Primary off-chain record	On-chain linkage
Purchase issuance	Loyalty transaction with purchase reference, partner ID, customer ID, and point amount.	TON transaction hash after token issuance.
Reward redemption	Redemption transaction and partner settlement status.	TON transfer, burn, or reference event.
Partner registration	Partner profile, tier, status, and KYC hash.	Partner registry smart-contract state.
Transaction history	Cached user-facing timeline for fast Telegram Mini App display.	Verifiable transaction hash when confirmed.

For purchase-based reward issuance, the backend validates cashier or POS permission, checks purchase uniqueness, confirms paid-order status, calculates points from the active campaign rule, creates a pending off-chain transaction, submits the TON issuance request, stores the transaction hash when available, and returns either pending or confirmed status to the client.

Deployment and acceptance checks

The deployment view consists of a Telegram Mini App client, Node.js backend services, PostgreSQL storage, and a TON blockchain adapter. The container-level configuration is omitted from the thesis because it is implementation-specific and does not add further analytical evidence. Instead, the acceptance checks below summarize how the architecture was evaluated at specification level.

Table 6: Architecture-level acceptance checks for the MVP.

ID	Check	Acceptance condition
AC-1	Telegram Mini App authentication	Backend accepts only valid Telegram init data and issues a short-lived API token.
AC-2	Purchase-based issuance	A confirmed paid order creates one off-chain transaction and one blockchain issuance request.
AC-3	Duplicate prevention	Repeated submission of the same purchase reference is rejected or returns the existing transaction state.
AC-4	Blockchain timeout handling	Delayed TON confirmation keeps the transaction pending and eligible for retry.
AC-5	Redemption control	Redemption fails when the available point balance is lower than the requested discount amount.
AC-6	Partner governance	Suspended or banned partners cannot issue or redeem loyalty points.
AC-7	Auditability	Transaction history displays type, points, status, timestamp, and blockchain transaction hash when available.

Stakeholder interview and survey summary

The stakeholder input was used during early requirement elicitation. It is presented in anonymized summary form because raw responses may contain personal or commercially sensitive information. The summary is not treated as statistically representative evidence; its role is to support design-stage requirement formation and to make the empirical basis of the MVP blueprint transparent.

Table 7: Anonymized stakeholder-input summary.

Stakeholder group	Input channel	Main recurring responses	Design implication
Confectionery owners and managers	Semi-structured interviews	Manual reconciliation is time-consuming; campaign rules should be adjustable; partner settlement must be auditable.	FR3, FR6, FR7 and the partner-governance model were prioritized.
Sales and cashier staff	Short operational interviews	Checkout interaction must remain fast; staff should not explain wallet mechanics during purchase.	NFR2 and the Telegram-centered low-friction interaction model were emphasized.
Customers	Short survey	In the 47-response survey, 66% reported current loyalty-program participation; users also value visible balances, simple redemption, and familiar mobile access.	The MVP focuses on balance, rewards, transaction history, and profile screens.
Partner businesses	Interview and requirements discussion	Cross-brand acceptance increases reward utility, but partner rights and liabilities must be controlled.	API interfaces, partner status control, and deployment boundaries were included.
Developers and operators	Architecture review discussion	Integration should use stable APIs, retry logic, and clear separation between on-chain and off-chain data.	Event-driven processing, PostgreSQL storage, and blockchain-adapter boundaries were selected.

These stakeholder inputs were mapped to the requirements and architectural artifacts discussed in Chapters 2–3: low-friction Telegram access, transparent balance history, partner settlement reliability, configurable campaigns, privacy protection through the hybrid data model, and failure tolerance through asynchronous confirmation and retry behavior.

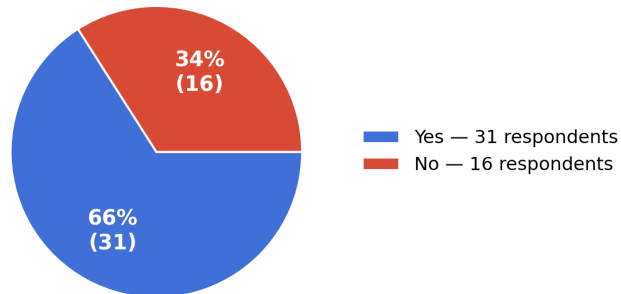
Current participation in loyalty/bonus programs (n = 47)

Figure 1: Anonymized survey-response evidence used for baseline loyalty-participation analysis.

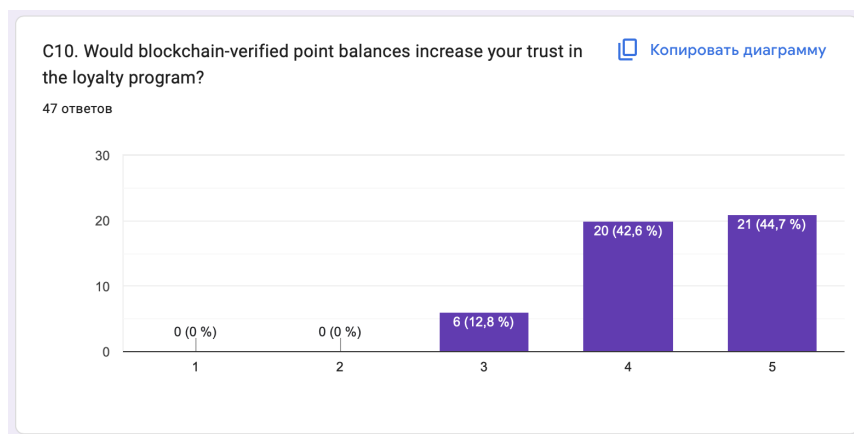


Figure 2: Anonymized survey-response evidence for perceived trust in blockchain-verified loyalty balances.